Journal of
Hydroinformatics

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Journal of Hydroinformatics Vol 27 No 3, 406 doi: 10.2166/hydro.2025.170

Evaluating the impact of different decomposition methods on the accuracy of reference evapotranspiration forecasts in humid regions

Zihuang Yan, Xianghui Lu* and Lifeng Wu

School of Soil and Water Conservation, Nanchang Institute of Technology, Nanchang 330099, China
*Corresponding author. E-mail: xianghuilu@nit.edu.cn

XL, 0000-0003-0638-3068

ABSTRACT

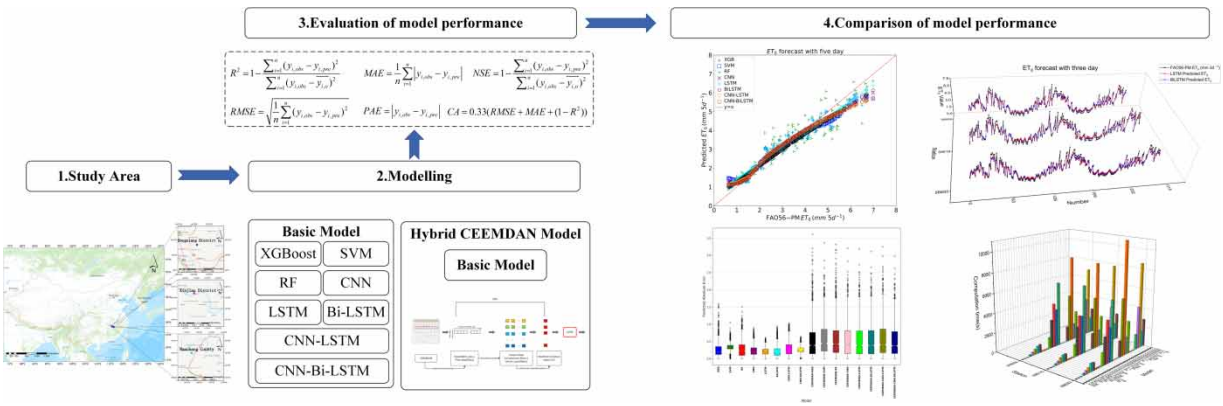
The forecasting of reference crop evapotranspiration (ET_0) plays a crucial role in irrigation scheduling. In this study, we employed deep learning (DL) models, including convolutional neural network (CNN), long short-term memory (LSTM), bidirectional LSTM (Bi-LSTM), CNN-LSTM, and CNN-Bi-LSTM, to forecast ET_0 at three different stations (Nanchang, Xinjian, and Dongxiang) in Jiangxi Province, China, for various forecast lead times (1, 3, 5, 7, 10, and 15 days). The experimental results demonstrate that LSTM and Bi-LSTM achieved the highest predictive accuracy across the three stations, with corresponding average values of coefficient of determination (R^2), mean absolute error (MAE), root mean square error (RMSE), Nash–Sutcliffe efficiency (NSE), and combined accuracy (CA) of 0.9646, 0.2456 mm day^{-1}, 0.2931 mm day^{-1}, 0.9646, and 0.2043, respectively. Compared with the baseline model, the hybrid models incorporating complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN), EEMD, and CEEMD exhibited inferior predictive performance. Additionally, their significantly longer runtime resulted in operational costs that were 8–10 times higher. In contrast, DL-based LSTM and Bi-LSTM are excellent algorithms for forecasting ET_0 across various time intervals.

Key words: CEEMD, CEEMDAN, deep learning, EEMD, LSTM, reference evapotranspiration

HIGHLIGHTS

- Accurately predicted the evapotranspiration under different cycles by deep learning LSTM and Bi-LSTM models based on the time-series structure.
- Revealed the potential of combining ensemble empirical mode decomposition, complete ensemble empirical mode decomposition, and complete ensemble empirical mode decomposition with adaptive noise with deep learning for evapotranspiration prediction.

GRAPHICAL ABSTRACT



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1. INTRODUCTION

Evapotranspiration (ET) is the physical process by which liquid water is transferred from the soil and plant surfaces into the atmosphere as water vapor. It is a crucial factor in the interactions among soil, plants, and the atmosphere, playing a significant role in the water balance and energy equilibrium at the Earth's surface (Allen *et al.* 1998; Liu *et al.* 2013). Moreover, ET_0 is an indispensable component in the terrestrial water, energy, and carbon cycles, making it crucial for water resource planning and management (Tao *et al.* 2018). To make irrigation systems work better, accurate predictions of crop evapotranspiration are needed. These predictions need to cover a long enough time in the future to take into account changes in water supply over time (Torres *et al.* 2011a). Similarly, accurate forecasting of crop water requirements (i.e., actual evapotranspiration, ET_a) is crucial for designing and planning irrigation scheduling (Perera *et al.* 2014). For large-scale irrigation system forecasting, Cetin *et al.* (2023) used the mapping evapotranspiration at high resolution and the internalized calibration (METRIC) model to estimate ET_0 using satellite and meteorological data, highlighting the importance of estimating ET_a for enhancing water efficiency.

Crop evapotranspiration (ET_a) can be estimated by multiplying the reference evapotranspiration (ET_0) with the crop coefficient (K_c). K_c represents the ratio between ET_a and ET_0 (Jensen 1968). Alternatively, ET_0 can be measured directly using an atmometer. However, this method is time-consuming and costly, making it less suitable for research purposes (Wright 1988; Tao *et al.* 2018). Simultaneously, research on ET_0 data is also highly noteworthy. Başakın *et al.* systematically summarized the process of estimating missing ET_0 data. From these studies, the complexity and nonlinearity of ET_0 data can be inferred (Başakın *et al.* 2023). Currently, the most well-known mathematical formula for determining ET_0 is the FAO-56 Penman–Monteith (PM) equation. This formula was proposed by the Food and Agriculture Organization of the United Nations (FAO) (Allen *et al.* 1998). The FAO-56 PM model, taking into account both thermodynamic and aerodynamic effects, is more accurate than existing empirical models. As a result, the FAO-56 PM method has been widely applied as a reference method in various regions and climates (Tao *et al.* 2018; Zhang *et al.* 2018; Granata 2019; Tikhamarine *et al.* 2019; Wu *et al.* 2019b; Lu *et al.* 2023). However, this model has a notable drawback. It requires a substantial amount of input meteorological data to be effectively utilized, including parameters such as maximum/minimum temperature, wind speed, relative humidity (RH), and solar radiation (Feng *et al.* 2017b). Therefore, the FAO-56 PM model can only be optimally employed when comprehensive meteorological data is available. For scenarios with limited meteorological data, it is advisable to use models that require fewer meteorological variables (Wen *et al.* 2015). In the past few decades, efforts have been made to address the issue of requiring extensive meteorological inputs by relying on empirical models (Mehdizadeh *et al.* 2017). Pendiuk *et al.* reduced the data requirements for estimating ET_0 by utilizing superconducting gravimeter data to calculate cumulative ET_0 (Pendiuk *et al.* 2023). In addition, many studies based on radiation, humidity, and temperature, and have aimed to estimate reference evapotranspiration using limited meteorological data (Xu & Singh 2002; Bickici Arikan *et al.* 2021). For example, there are temperature-based methods such as the Hargreaves and the modified Hargreaves methods (Hargreaves & Samani 1985; Luo *et al.* 2014), radiation-based methods including the Priestley–Taylor, Marking, and Ritchie models (Priestley & Taylor 1972; Feng *et al.* 2016), as well as the mass-transfer-based Trabert model (Trabert 1896). Additionally, these models are more suitable for forecasting ET_0 on weekly and monthly cycles and may not be as well-suited for daily ET_0 forecasting (Torres *et al.* 2011a). We employ Chen's nonlinear theoretical model within a dual-source trapezoid framework based on dry/wet edges to estimate land surface ET. This model enhances the estimation of daily-scale ET_0, thereby improving the prediction accuracy of regional evapotranspiration and its components. This research is closer to a physical model for ET_0 prediction (Chen *et al.* 2023). These models involve complex physical processes, and predictive accuracy still has room for improvement.

In contrast to attempts with empirical models, researchers have sought to address the challenges associated with empirical models by employing data-driven artificial intelligence (AI) and machine learning (ML) methods to establish mathematical models for forecasting ET_0. Due to the high nonlinearity of meteorological data, traditional empirical models struggle to accurately capture the relationships between meteorological variables, thereby hindering their ability to forecast ET_0 effectively. However, ML does not require a clear understanding of data variables in advance and can offer simple solutions to nonlinear and multivariate problems (Kisi 2015; Wang *et al.* 2017). Due to its capability to operate without prior knowledge of the underlying data and physical processes and its ability to handle nonlinearity, ML becomes robust, efficient, and reliable in problem-solving and decision-making (Goyal *et al.* 2023). In past research, various ML techniques have been applied to forecast ET_0. For instance, Ferreira *et al.* used artificial neural networks (ANNs) and support vector machines (SVMs) with

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limited meteorological data to forecast ET_0 at 203 stations in Brazil. The results indicated that ANN and SVM exhibited good forecasting performance (Ferreira *et al.* 2019). Vosoughifar *et al.* established ET_0 equations for coastal areas using multi-adaptive regression splines (MARS) and genetic expression programming (GEP), discovering they possess a more accurate estimation capability than traditional regression-based ET_0 formulas (Vosoughifar *et al.* 2023). In addition to that, other models such as M5 model trees (Pal & Deswal 2009; Rahimikhoob 2014; Kisi & Kilic 2016), random forest (RF) (Feng *et al.* 2017a), gradient boosting decision tree (GBDT) (Fan *et al.* 2018; Wu *et al.* 2019a), and ANN-based models like multi-layer perceptron (MLP) (Torres *et al.* 2011a; Ladlani *et al.* 2014; Traore *et al.* 2016), generalized regression neural network (KiŞI 2006), radial basis function neural network (RBFNN) (Trajkovic 2005; Ladlani *et al.* 2012), and extreme learning machine (ELM) (Abdullah *et al.* 2015; Gocic *et al.* 2016) have been employed to forecast ET_0. In previous studies, to further improve the forecasting accuracy of ET_0 within the framework of ML, bio-inspired optimization algorithms have been incorporated. For example, in the forecasting of ET_0 for arid and humid regions in China, a hybrid model combining the whale optimization algorithm (WOA) with extreme gradient boosting (XGBoost) was employed. The results indicated that this strategy could provide a more accurate estimation of daily evaporation (Yan *et al.* 2021). Wu *et al.* utilized four bio-inspired optimization algorithms – ant colony optimization, genetic algorithm, flower pollination algorithm, and cuckoo search algorithm – to optimize the ELM model for daily ET_0 forecasting. The results indicated that the ELM-FPA and ELM-CAS models outperformed the base ELM model regarding forecasting accuracy (Wu *et al.* 2019b). Zhao *et al.* (2022) combined two bio-inspired optimization algorithms, golden eagle optimization (GEO) and sparrow search algorithm (SSA), with ELM. The ELM model with the SSA was considered the optimal model for forecasting ET_0.

While conventional ML models have made significant strides in the daily forecasting of ET_0, there are still limitations to these theoretical models when it comes to forecasting ET_0 with higher nonlinearity, larger datasets, and more complex cycles. In recent years, deep learning (DL) has been applied to various fields, and its performance has surpassed that of traditional ML models (Shi *et al.* 2018; Ghimire *et al.* 2019; Du *et al.* 2021; Luo *et al.* 2021; Yan *et al.* 2023). Furthermore, a growing body of research employs DL for the time-series forecasting of ET_0. This differs from traditional forecasting methods, as time-series forecasting involves forecasting a sequence using a sequence, distinct from the conventional regression task where independent variables are used to forecast dependent variables. Time-series forecasting also provides a better understanding of the forecasted trends and aligns well with real-world variations. Sabanci *et al.* (2023) predicted ET_0 for 12 stations with variable climatic characteristics in the Central Anatolia region (CAR) using long short-term memory (LSTM), ANN, and MARS models. The results indicated that LSTM, ANN, and ML-based MARS showed high accuracy in estimating ET_0. Yin *et al.* (2020) forecasted daily ET_0 based on a limited set of meteorological variables using a hybrid bidirectional LSTM (Bi-LSTM) model. The results indicated that the Bi-LSTM model yielded the best ET_0 forecasting for all stations. Saggi & Jain (2019) estimated reference evapotranspiration in the northern region of Punjab, India, using a DL-MLP in comparison with traditional ML models such as generalized linear model (GLM), RF, and gradient boosting machine (GBM). The results indicated that the DL model achieved the best forecasting performance. Additionally, by combining DL models with convolutional neural networks (CNNs), one can further enhance the training of DL models through the convolutional calculations of CNN. For instance, Ferreira & da Cunha (2020a) used a hybrid CNN-LSTM model to forecast lagged ET_0, combining a one-dimensional CNN with an LSTM model. The results suggest that CNN-LSTM performs slightly better than RF and ANN but does not demonstrate the highest accuracy. In addition, combining ML with metaheuristic algorithms to optimize model parameters is also a common strategy. For example, Yuan *et al.* optimized the parameters of an LSTM model using the ant lion optimization (ALO) algorithm (LSTM-ALO) for monthly runoff prediction (Yuan *et al.* 2018). Ikram *et al.* utilized the reptile search algorithm and the weighted vector optimizer to enhance the performance of the LSTM model (LSTM-INFO) in predicting water temperature (Ikram *et al.* 2023). The application of DL in forecasting ET_0 still requires further analysis and exploration to maximize its inherent value.

Despite extensive studies and proven effectiveness of ML and DL models in forecasting ET_0, bias toward data dependency persists. If ET_0's increased nonlinearity, uncertainty, and randomness fail to capture relevant features/patterns in the time series, including trends, periodicity, seasonality, anomalies, and abrupt change peaks, it becomes increasingly difficult for DL to produce accurate forecasts (Ali *et al.* 2023). By decomposing the original data to obtain signal components and forecasting each signal component separately, even though there are multiple features in actual problems, each feature undergoes one decomposition at a time (Prasad *et al.* 2019). The decomposed signals become smoother with more robust periodicity, making them better suited for DL forecasting. Ali *et al.* employed the multivariate variational mode decomposition (MVMD) and monotonic empirical mode decomposition (MEMD) to decompose the original data. They then used a

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combination of RF, convolutional feedforward neural network (CFNN), ELM, and boosted regression trees (BRT) to forecast daily ET_0. The results indicated that MVMD-BRT, compared with the baseline model (BRT), could provide more accurate daily ET_0 forecasting accuracy (Ali *et al.* 2023). Zheng *et al.* employed the MVMD method combined with the soft feature filter (SoFeFilter) and gated recurrent unit (GRU) model to predict ET_0 for $t+1$ day. Experimental results demonstrated that at the Gympie station, the MVMD-SoFeFilter-GRU model achieved the highest performance metrics. These findings confirm that the MVMD-SoFeFilter-GRU model provides the most accurate predictions for ET_0 1 day ahead (Zheng *et al.* 2023). Similarly, by integrating LSTM with the MEMD method and the Boruta-RF algorithm, ET was forecasted for drought-prone areas in Queensland, Australia. The results indicated that the MEMD-Boruta-LSTM hybrid model outperformed standalone models such as deep neural network (DNN), decision tree (DT), and LSTM, achieving the highest predictive performance across all study locations (Jayasinghe *et al.* 2021). In addition, time-varying correlation decomposition methods have been applied to ET_0 forecasting. Karbasi *et al.* employed the time-varying filter-based empirical mode decomposition (TVF-EMD) technique, coupled with the partial autocorrelation function (PACF), to calculate significant lag values from the decomposed subsequences (i.e., IMFs). The extra tree-Boruta feature selection algorithm was then used to extract key features from the IMFs. By integrating TVF-EMD with the bidirectional recurrent neural network (Bi-RNN), MLP, RF, and XGBoost, weekly evaporation was predicted and evaluated using various performance metrics. The results demonstrated that, compared to models without decomposition, the TVF-EMD hybrid ML models achieved superior forecasting accuracy for weekly ET_0 at the Redcliffe and Gold Coast stations (Karbasi *et al.* 2023). Lu *et al.* combined the variational mode decomposition (VMD) algorithm with the backpropagation neural network (BPNN) model. The results showed that the VMD-BPNN hybrid forecasting model outperformed the empirical mode decomposition (EMD) algorithm with the BPNN (EMD-BPNN) and the ensemble empirical mode decomposition (EEMD) algorithm with BPNN (EEMD-BPNN) models. Additionally, the forecasting accuracy of the VMD-BPNN model was significantly higher than that of single models such as BPNN, support vector regression (SVR), and gradient boosting regression tree (GBRT) (Lu *et al.* 2023). Commonly used decomposition algorithms include Fourier spectrum analysis (Soman *et al.* 2015) and discrete wavelet transform (Mallat 1989). Karbasi *et al.* proposed a time-series decomposition technique known as empirical Fourier decomposition (EFD). Autocorrelation analysis was employed to determine significant lag values, and EFD was applied to decompose the ET_0 data. Based on the EFD results, lagged data were created, and the K-best feature selection algorithm was used to identify key features. Models were evaluated using the correlation coefficient (R) and root mean square error (RMSE). The results demonstrated that incorporating EFD-decomposed ET_0 data significantly enhanced the predictive performance of ML models (Karbasi *et al.* 2024). Katipoğlu estimated monthly ET values for the Hakkari Province by combining SVM, bagging trees, and boosting trees with wavelet transformation. Model performance was evaluated using RMSE, MAE, the coefficient of determination (R^2), and Taylor diagrams. The results indicated that hybrid wavelet-ML models, constructed by decomposing input combinations into subcomponents via wavelet transformation, generally produced more accurate predictions compared with standalone ML models (Katipoğlu 2023). Additionally, a method different from the decompositions mentioned above is complete ensemble empirical mode decomposition adaptive noise (CEEMDAN) (Yeh *et al.* 2010), which has been applied to many other fields with satisfactory results (Zhang *et al.* 2017; Gao *et al.* 2020; Yan *et al.* 2023). It is generally believed that decomposition methods have advantages, but in geographically complex environments, where numerous influencing factors and frequent meteorological changes occur, time series-based methods can be risky. This study verifies this risk from multiple perspectives. However, based on the author's understanding, there is currently no specific analysis of the application of CEEMDAN combined with CNN and DL for forecasting ET_0. A detailed analysis of this algorithm is of significant reference value for future research on ET_0 forecasting.

Due to the nonlinearity among climate variables, accurately predicting ET_0 has become a challenge. Related literature shows few studies on DL-based models combined with CEEMDAN models to forecast ET_0 in China. This study proposes four new methods for forecasting ET_0 using CNN, LSTM, Bi-LSTM, and CNN-Bi-LSTM combined with CEEMDAN. This study compared hybrid models that integrate different decomposition algorithms, including EEMD, CEEMD, and CEEMDAN, with standalone model approaches, offering a more accurate forecasting strategy for multi-period ET_0 prediction. Therefore, the objectives of this study are as follows: (1) Determine the importance and forecasting potential of CEEMDAN combined with base models XGBoost, SVM, RF, CNN, LSTM, Bi-LSTM, CNN-LSTM, and CNN-Bi-LSTM for ET_0 forecasting; (2) Develop the base model and CEEMDAN models for forecasting ET_0 at different periods; (3) Compare the accuracy, stability, and computational cost of baseline models with models based on EEMD, CEEMD, and CEEMDAN for predicting ET_0 over different periods, and identify the optimal model; (4) Analyze the compatibility of the base model with the

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combination of CEEMDAN and CNN models for predicting ET_0 at different periods, aiming to explore potential synergies and optimize overall predictive performance.

2. METHODOLOGY

2.1. Study area and data collection

The meteorological data for this study were collected from three different weather stations in Jiangxi Province, China (Figure 1). Three weather stations were selected to assess the model's generalizability for forecasting ET_0 in this province: Nanchang, Xinjian, and Dongxiang in Jiangxi Province (Table 1). The research data used in this study were derived from daily maximum temperature (T_{max}), daily minimum temperature (T_{min}), sunshine duration (SSD), wind speed (Wind Speed), daily average RH, and annual precipitation (P) from the China Meteorological Administration website (www.cma.com) for the period from 1 September 2001 to 31 August 2021. Table 1 describes the average values of each input meteorological variable and the target variable ET_0 for the three stations. It also provides specific information about the

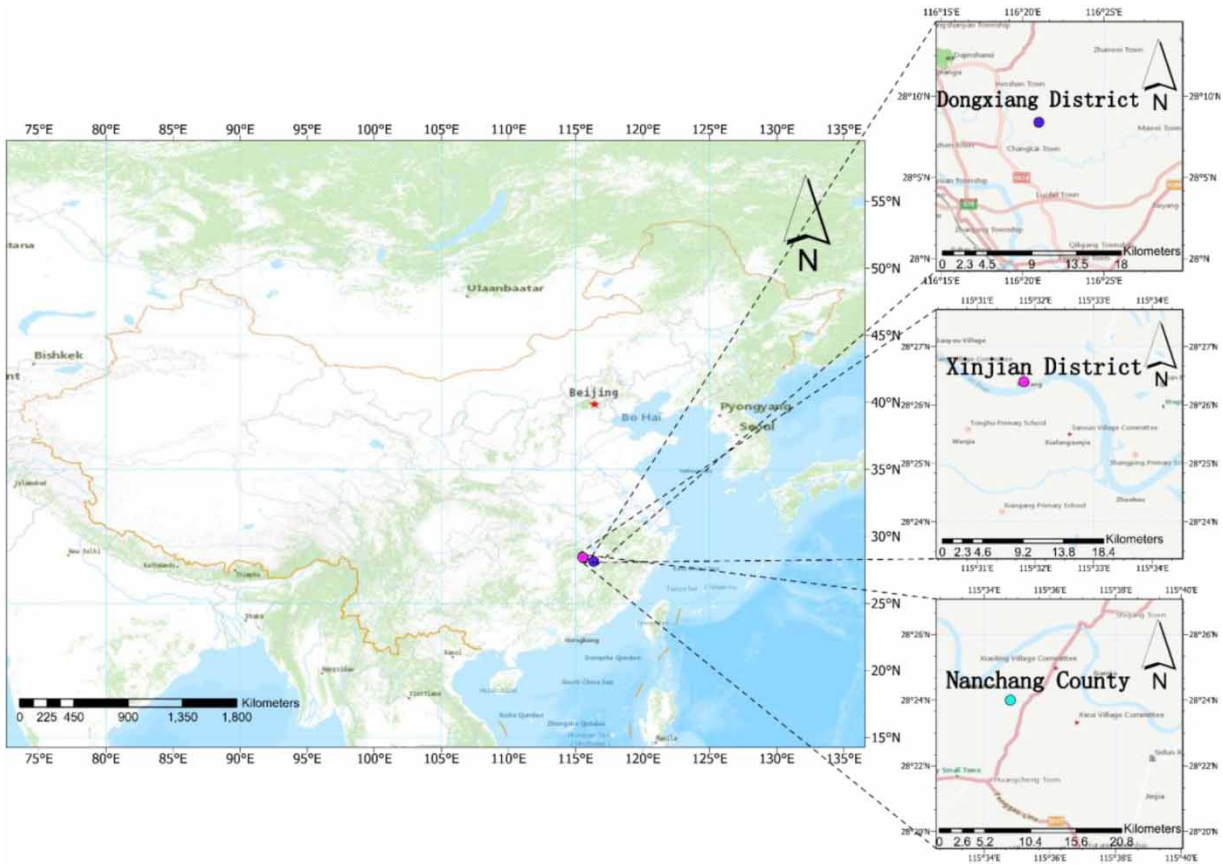

Figure 1 | Map of the three study stations in this paper.

Table 1 | The geographical location information and the average values of meteorological variables for the three study stations in this research

Station name	Station ID	Longitude (E)	Latitude (N)	Altitude (m)	T_{max} (°C)	T_{min} (°C)	SSD (h)	Wind (ms^{-1})	RH (%)	P ($mmyear^{-1}$)	ET_0 ($mmday^{-1}$)
Nanchang	58606	115.58	28.40	46.9	22.77	15.85	5.07	1.81	72.72	1,627.6	3.00
Xinjian	58693	115.53	28.44	40	22.97	16.23	4.64	1.35	70.08	1,546.4	2.86
Dongxiang	58618	116.35	28.14	50.6	23.19	15.32	4.45	2.36	77.78	1,823.3	2.89

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geographical location of each station, including latitude, longitude, station ID, and elevation. This fundamental analysis is a reference for understanding the meteorological data at the three stations. Additionally, before using the data as input for the model, we performed min–max normalization, constraining the values to be within the range of 0–1. This normalization enhances the model's performance. In this experiment, the dataset was split into a training set and a test set in an 8:2 ratio, and predictions were made on the test set.

2.2. FAO-56 PM model

The FAO-56 PM equation has been widely used as a reference equation for estimating ET_0 in various scientific fields, such as agriculture and hydrology (Fan *et al.* 2018; Chu *et al.* 2019; Jerin *et al.* 2021; Zhang *et al.* 2023). In this study, the daily ET_0 calculated by the PM equation is used as the target value for evaluating the performance of each model. The mathematical expression of the PM equation is as follows:

$$ET_0 = \frac{0.408\Delta(R_n - G) + \gamma\frac{900}{T+273}U_2(e_s - e_a)}{\Delta + \gamma(1 + 0.34U_2)} \tag{1}$$

where ET_0 represents reference evapotranspiration (mm day^{-1}); Δ is the slope of the vapor pressure curve (kPa °C^{-1}); R_n is the net solar radiation at the Earth's surface (MJ m^{-2} day^{-1}); G is the soil heat flux (MJ m^{-2} day^{-1}); γ is the psychrometric constant (kPa °C^{-1}); T is the average air temperature at a height of 2 m (°C); U_2 is the wind speed at a height of 2 m (m s^{-1}); e_s and e_a are the saturation vapor pressure (kPa) and the actual vapor pressure (kPa), respectively.

2.3. Conventional ML models for forecasting ET_0

Researchers have widely applied conventional ML models in ET_0 forecasting, yielding excellent forecasting results (Fan *et al.* 2018; Wu *et al.* 2019a; Ferreira & da Cunha 2020b; Yan *et al.* 2021). In previous studies, XGBoost (Fan *et al.* 2018), SVM (Granata 2019), and RF (Ali *et al.* 2023) have demonstrated stable performance in ET_0 forecasting. This study used XGBoost, SVM, and RF as baseline models to compare with DL models, thereby enhancing the rigor of the DL forecasting results.

2.3.1. Extreme gradient boosting

XGBoost is an algorithm or engineering implementation based on GBDT proposed by Chen & Guestrin (2016). The basic idea of XGBoost is similar to GBDT, but XGBoost introduces the concept of 'boosting', aggregating forecasting from 'weak' learners to generate a strong learner. Additionally, XGBoost incorporates second-order derivative optimization, making the loss function more precise. The regularization term optimization helps prevent the overfitting of trees. In addition, XGBoost is an efficient distributed implementation, making its training speed faster.

2.3.2. Support vector machine

SVM is a supervised ML algorithm proposed by Cortes & Vapnik (1995). Based on the principle of structural risk minimization (SRM), SVMs seek to minimize the upper bound of generalization error rather than the upper bound of empirical error. To make predictions in problems that are not linearly related, SVM models can use a set of high-dimensional linear functions to make regression functions (Pai & Lin 2005). SVM employs different kernel functions as a key component to address non-linear problems. Kernel functions can transform the original features of the data from a lower-dimensional space to a higher-dimensional feature space, making the data linearly separable in the new space. This study utilized the RBF kernel function because, compared with other kernel functions, it demonstrated better performance in forecasting ET_0 (Kisi 2015).

2.3.3. Random forest

RF was introduced by Breiman (2001). An ensemble learning method utilizes the 'bagging' concept to integrate a collection of decision trees with controlled variance. By combining multiple decision trees, RF aims to enhance the accuracy of the model. The ensemble nature of RF, which uses the voting results of multiple decision trees, makes it more robust than a single DT and better able to handle overfitting problems. RF randomly selects the best factor from a subset of forecasters at each node to split it, thereby improving forecasting performance (Liaw & Wiener 2002).

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2.4. DL models for forecasting ET_0

This paper employs time series models, including CNN, LSTM, and Bi-LSTM, which are capable of capturing complex linear relationships in time series data. In this study, multi-period ET_0 forecasting was conducted, where LSTM better captures long-term temporal dependencies.

2.4.1. Convolutional neural network

CNN has been applied to various fields to address problems, yielding favorable results (Cao *et al.* 2018; Hoseinzade & Haratizadeh 2019; Viton *et al.* 2020). Traditional CNN structures consist of three different types of essential layers: convolutional layers, pooling layers, and fully connected layers. For feature extraction, CNN can automatically extract features from input data, which is very different from manually extracting features, and it eliminates the need for data preprocessing (Ferreira & da Cunha 2020b). Moreover, CNN has the characteristic of parameter sharing, where weight parameters are shared at different locations, reducing model complexity, training time, and memory requirements and enhancing robustness. Since CNN is commonly used for image processing problems, it employs 2D convolutional filtering, as images are typically composed of two-dimensional arrays (matrices) (Ferreira & da Cunha 2020a). However, when dealing with time series-based problems in this experiment, CNN with one-dimensional (1D) convolutional filters was used.

2.4.2. Long short-term memory

LSTM is an RNN architecture proposed by Hochreiter & Schmidhuber (1997). DL models, including LSTM, have been shown to outperform shallow ML models and traditional data analysis methods (Janiesch *et al.* 2021). LSTM addresses the vanishing and exploding gradient problems encountered during backpropagation in time-series forecasting tasks, which were issues associated with traditional RNNs. LSTM introduces a gating mechanism, and its basic unit consists of a memory cell and three adaptive multiplicative gates (input gate, update gate, and forget gate). LSTM primarily operates by multiplying the cell's input with the input gate, selectively forgetting information in the cell state through the forget gate, and ultimately obtaining the next state value by multiplying the activation with the output gate. The mathematical expression for the LSTM model is as follows:

$$f_t = \sigma(w_{\mathrm{f}} \cdot [h_{t-1}, x_t] + b_{\mathrm{f}}) \tag{2}$$

$$i_t = \sigma(w_{\mathrm{i}} \cdot [h_t - 1, x_t] + b_{\mathrm{i}}) \tag{3}$$

$$\widetilde{C}_t = \tanh(w_{\mathrm{c}} \cdot [h_t - 1, x_t] + b_{\mathrm{c}}) \tag{4}$$

$$o_t = \sigma(w_{\mathrm{o}} \cdot [h_t - 1, x_t] + b_{\mathrm{o}}) \tag{5}$$

$$C_t = f_t \cdot C_{t-1} + \widetilde{C}_t \cdot i_t \tag{6}$$

$$h_t = o_t \cdot \tanh(C_t) \tag{7}$$

where σ is the activation function for the LSTM forget gate, input gate, and output gate, with a range from 0 to 1. Tanh is the activation function for the update unit, ranging from -1 to 1. The activation functions control the output of each link. w_{f}, w_{i}, w_{c}, and w_{o} represent the weight matrices for the forget gate, input gate, update unit, and output gate, respectively. b_{f}, b_{i}, b_{c}, and b_{o} represent the bias terms in each of the function expressions. h_t represents the output from the output gate o_t to the next memory cell.

2.4.3. Bidirectional long short-term memory

Bi-LSTM is a Bi-RNN introduced by Schuster & Paliwal (1997). Bi-RNN can be trained to forecast only within a predefined future, where each training sequence has both forward and backward RNNs. In such continuous time series tasks, Bi-LSTM can simultaneously capture information from the past and the future. Unlike LSTM, the hidden state h_t in LSTM cannot capture information from the future. The basic structure of Bi-LSTM consists of two LSTMs, where one LSTM is responsible for training on sequentially collected input data from the front, and the other is responsible for training on data collected from the back. This architecture enhances the model's accuracy and performance in handling time series tasks. The hidden layer and

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the final input of Bi-LSTM can be represented as:

$$h_t = \mathrm{LSTM}(x_t, h_{t-1}) \tag{8}$$

$$h_i = \mathrm{LSTM}(x_t, h_{i-1}) \tag{9}$$

$$y_t = \sigma(\alpha_t h_t + \beta_t h_i + C_t) \tag{10}$$

where LSTM is the standard model, y_t is the final output, α_t and β_t represent the forward and backward output weights of the hidden layer, and C_t is the bias term of the hidden layer at time t.

2.4.4. CNN-LSTM and CNN-Bi-LSTM

CNN-LSTM and CNN-Bi-LSTM combine the advantages of CNN with LSTM and Bi-LSTM, making the models more stable. The process adopted in this study for CNN-LSTM and CNN-Bi-LSTM involves taking the previous time series as input data and maintaining consistency in input dimensions with 1D-CNN. The input, convolution, and pooling layers are retained to extract features. The extracted features are then used as input for LSTM and Bi-LSTM, and the final result is forecasted through LSTM and Bi-LSTM. The specific mathematical expressions are as follows:

$$y_{i,\mathrm{cnn}} = \mathrm{CNN}(x_i) \tag{11}$$

$$y_{i,\mathrm{lstm}} = \mathrm{LSTM}(y_{i,\mathrm{cnn}}) \tag{12}$$

$$y_{i,\mathrm{bi-lstm}} = \mathrm{Bi-LSTM}(y_{i,\mathrm{cnn}}) \tag{13}$$

where x_i represents the time series input to CNN, $y_{i,\mathrm{cnn}}$ represents the features after processing by CNN; $y_{i,\mathrm{lstm}}$ and $y_{i,bi-lstm}$ represent the forecasted output results of CNN-LSTM and CNN-Bi-LSTM, respectively. The structure diagrams of CNN-LSTM and CNN-Bi-LSTM are shown in Figure 2.

2.5. Structure of the designed CEEMDAN mixed DL model

2.5.1. Complete ensemble empirical mode decomposition adaptive noise

This paper uses the decomposition algorithm CEEMDAN, developed by Torres *et al.*, based on the EEMD algorithm (Torres *et al.* 2011b). The EMD and EEMD algorithms share the common feature of decomposing the original signal into high-frequency signals called intrinsic mode functions (IMFs) after multiple iterations. EMD and EEMD algorithms can alleviate mode mixing issues by adding positive and negative Gaussian noise to the decomposed signals. This characteristic is also present in CEEMDAN. However, both of these algorithms inevitably leave some white noise in the obtained IMFs

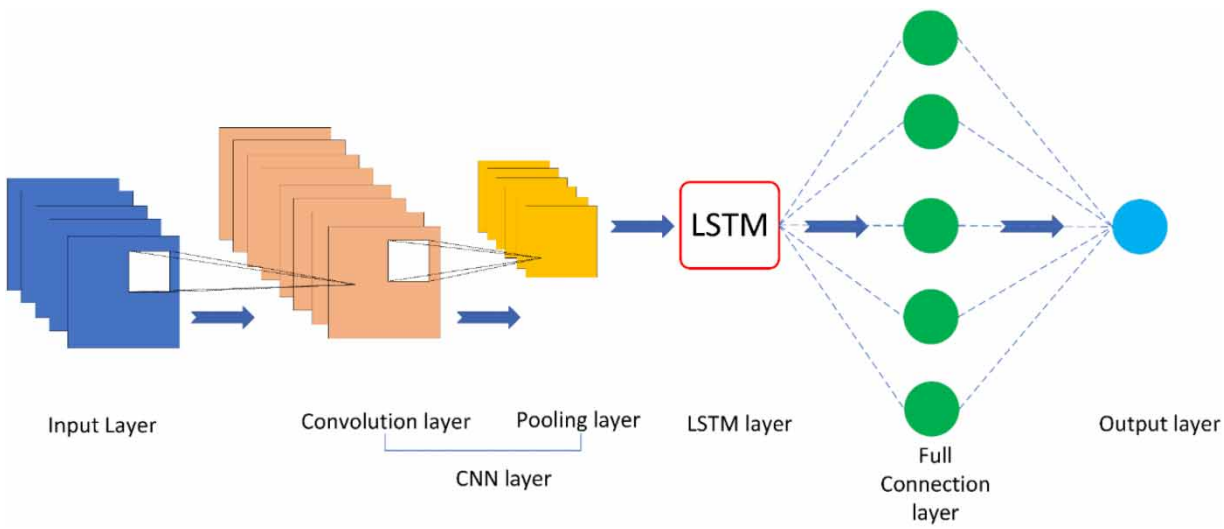

Figure 2 | CNN-LSTM structure diagram, CNN-Bi-LSTM structure only needs to replace the LSTM layer part to Bi-LSTM.

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from the decomposition of the original signal, which can impact the model's forecasting performance. CEEMDAN, on the other hand, differs in that its key advantage lies in its adaptability. It can automatically adjust noise estimation based on the signal's local characteristics, thereby improving signal decomposition's accuracy and stability. The implementation process of CEEMDAN is as follows:

(1). Add white noise $\omega^i(t)$ with a standard deviation of ε to the original signal $X^i(t)$:

$$X^i(t) = X(t) + \varepsilon_0 \omega^i(t), \quad i = 1, 2, \ldots, k \tag{14}$$

where k represents a real number.

(2). Decompose the signal set using EMD, then average each decomposition's combined parts.

$$\overline{\mathrm{IMF}_1}(t) = K^{-1} \sum_{i=1}^{K} \mathrm{IMF}_1^i(t) \tag{15}$$

(3). Calculate the first-order residuals.

$$r_1(t) = X(t) - \overline{\mathrm{IMF}_1}(t) \tag{16}$$

(4). Further, decompose the signal $r_1(t) + \varepsilon_1 \mathrm{EMD}_1(\omega^i(n))$ using EMD and calculate the second IMF mode.

$$\overline{\mathrm{IMF}_2}(t) = K^{-1} \sum_{i=1}^{K} \mathrm{EMD}_1(r_1(t) + \varepsilon_1 \mathrm{EMD}_1(\omega^i(t))) \tag{17}$$

where $\mathrm{EMD}_k(\cdot)$ represents the $k-$th IMF mode component obtained by EMD.

(5). In the next stage, the $k+1-$th component and $k-$th residue are calculated, as given by the following equations:

$$\overline{\mathrm{IMF}_{k+1}}(t) = K^{-1} \sum_{i=1}^{K} \mathrm{EMD}_1(r_k(t) + \varepsilon_k \mathrm{EMD}_k(\omega^i(t))) \tag{18}$$

$$r_k(t) = r_{k-1}(t) - \overline{\mathrm{IMF}_k}(t) \tag{19}$$

(6). Repeat Equations (18) and (19) until the residual component r_k no longer satisfies the decomposition conditions, and the decomposition stops. Finally, the original signal $X(t)$ can be represented as:

$$X(t) = \sum_{i=1}^{K} \overline{\mathrm{IMF}_i}(t) + R(t) \tag{20}$$

where $R(t)$ denotes the final residuals.

CEEMDAN is commonly used for data decomposition and denoising, and its reasonable application to various models is crucial. The specific process is illustrated in Figure 3.

2.5.2. Ensemble empirical mode decomposition and complete ensemble empirical mode decomposition

To further validate the robustness and complexity analysis of CEEMDAN decomposition, this study proposes a comparison with hybrid ML methods based on EEMD and CEEMD. EEMD is an improvement on the original EMD. EEMD introduces multiple iterations of white noise into the original signal, performs EMD on each noise-added signal, and averages the results. This approach effectively mitigates the mode mixing problem, enhancing the robustness and accuracy of the decomposition (Wu & Huang 2009). CEEMD further refines EEMD. While EEMD reduces mode mixing, residual noise can still occur in some cases. CEEMD addresses this issue by introducing symmetrically distributed positive and negative white noise into the signal, completely eliminating residual noise and ensuring the completeness of the decomposition (Torres *et al.* 2011a). The

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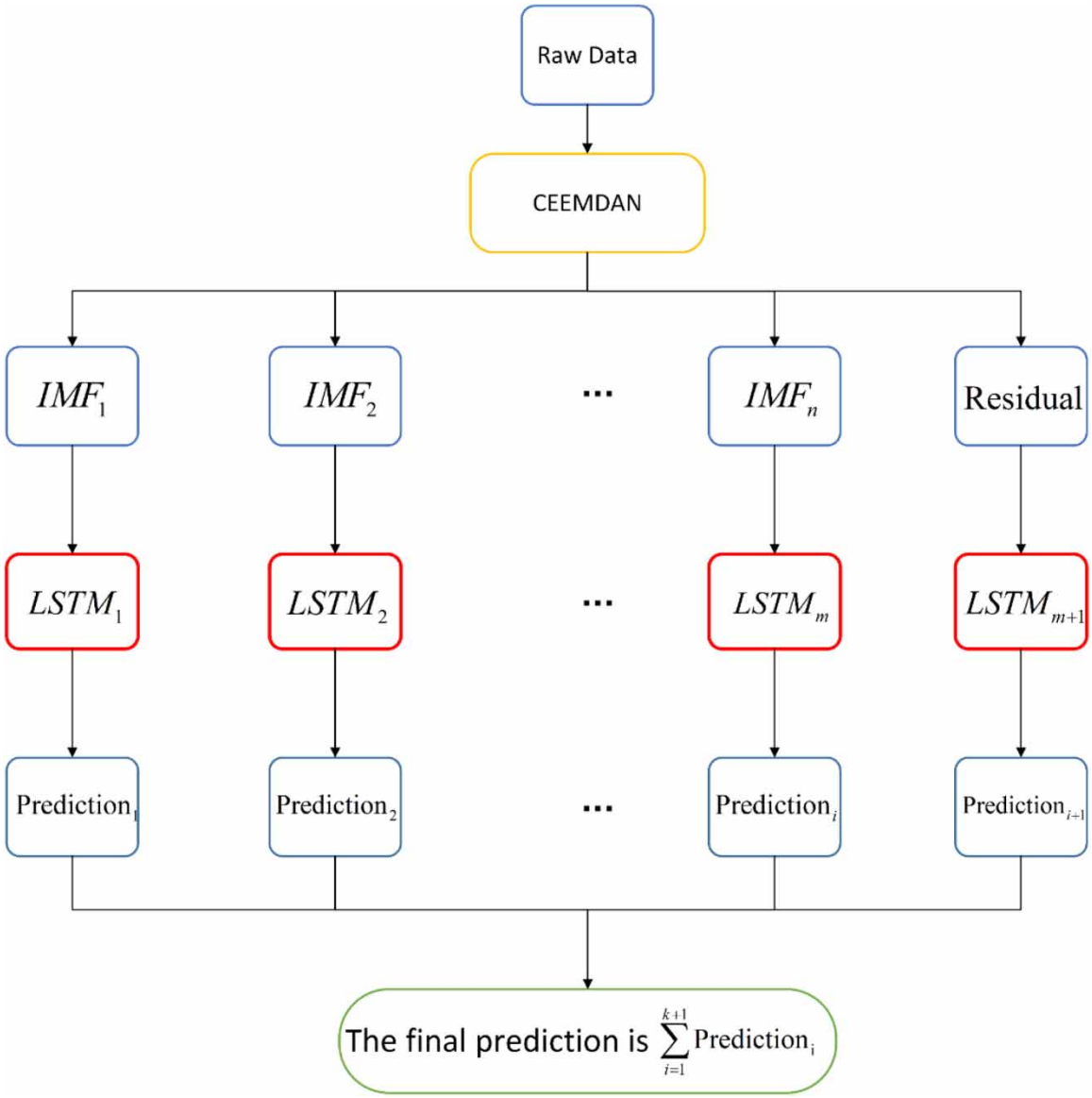

Figure 3 | Flowchart of the CEEMDAN-LSTM forecasting process. Note that the LSTM part can be replaced with other models for ET_0 forecasting.

forecast structures of hybrid ML models for ET_0 based on EEMD and CEEMD algorithms are consistent with the structure shown in Figure 3. By replacing the CEEMDAN components with EEMD or CEEMD, the corresponding schematic diagram of the hybrid ML models for ET_0 prediction is obtained.

2.5.3. CEEMDAN-CNN-LSTM and CEEMDAN-CNN-Bi-LSTM

The hybrid models CEEMDAN-CNN-LSTM and CEEMDAN-CNN-Bi-LSTM involve decomposing the original signal using CEEMDAN. The results are then used as input for CNN to extract features further. The final forecasting is obtained using the features processed by CNN as input for LSTM and Bi-LSTM. For CEEMDAN-CNN-LSTM and CEEMDAN-CNN-Bi-LSTM, the process is shown in Figure 4 and further described as follows.

2.5.4. Forecasting strategy: ET_0 periodic decomposition and sliding window

The forecasting structure adopted in this paper involves using the sliding window algorithm to sum and average daily ET_0 over multiple days, creating different periods of ET_0. The sliding window algorithm has been widely applied in time series

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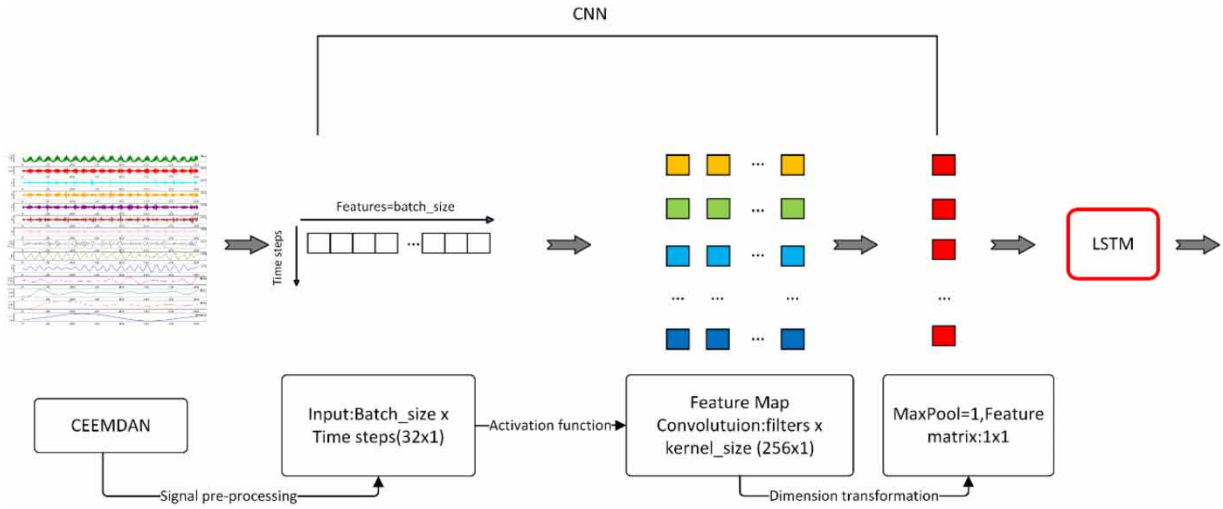

Figure 4 | CEEMDAN-CNN-LSTM structure. Note that LSTM can be replaced by Bi-LSTM that is the structure of CEEMDAN-CNN-Bi-LSTM. Note that replacing CEEMDAN with EEMD or CEEMD results in the corresponding EEMD-CNN-LSTM or CEEMD-CNN-LSTM structure.

tasks for processing (Yahmed *et al.* 2015; Chou & Ngo 2016; Carmona-Cabezas *et al.* 2019). It involves setting a fixed-size window as a moving step to construct the time series used as input for the model. In this paper, the forecasting of ET_0 is done by forecasting the following time series using the previous one. Additionally, the sliding window is used to forecast ET_0 in each sequence, and the results are compared with the actual values. In this study, daily ET_0 serves as the foundation. The values for ET_0 for each period of 3, 5, 7, 10, and 15 days are obtained by adding up the daily values and calculating the respective averages. These averages represent the ET_0 for periods of 3, 5, 7, 10, and 15 days. The specific forecasting structure is shown in Figure 5. We defined a sliding time window with a length of 1, and the model's input time series is constructed using the sliding window algorithm.

2.6. Tuning of model parameters

This study employed the grid search (GS) hyperparameter optimization method. Since the parameter space of ML and DL algorithms may consist of actual values or have infinite space, GS involves setting multiple values for each parameter within defined bounds. This approach is beneficial for parallelization, making it easy to explore the parameter space and find the optimal model parameters (Liashchynskyi & Liashchynskyi 2019). The ML models, XGBoost, SVM, and RF, have relatively few model parameters, so hyperparameter optimization was not performed on them. XGBoost chose the reg:squarederror parameter, commonly used in regression tasks, as the objective function. The RBF kernel function generally has good generalization ability and is increasingly used in nonlinear mapping SVM (Han *et al.* 2012). RBF was chosen as the kernel function for SVM. The number of decision trees (N_Estimators) for RF and XGBoost are 100 and 42, respectively. RF requires more decision trees to reduce prediction bias, while XGBoost, with its gradient boosting technique, improves each tree based on the previous one. Therefore, XGBoost uses fewer trees (N_Estimators = 42) for training. In the RF model, random_state serves as the random seed to ensure result reproducibility. In this study, setting it to 50 achieved better forecasting results. In the RF model, setting n_jobs to 1 uses only one computing core, saving computational resources and avoiding interference with other tasks. Setting oob_score = True provides an additional performance evaluation metric, which is useful for tuning model parameters. Setting max_features = 1 ensures that each tree uses all features. min_samples_leaf = 10 specifies that each leaf node must contain at least 10 samples to prevent overfitting. DL models have more parameters. The best model parameters for the DL parameters were determined through GS. In CNN, filters and kernel_size refer to the number of convolutional kernels used for feature extraction and the size of the 1×1 kernels, respectively. In LSTM and Bi-LSTM models, the unit parameter represents the number of neurons in the LSTM layer. Higher units increase the model's learning capacity and complexity, but they also lead to greater computational demands and training time, as well as a higher risk of overfitting (Yu *et al.* 2019). Through multiple experiments and grid searches, this study determined that 150 units is the most suitable choice. Dropout = 0.1 means that each neuron has a 10% chance of being dropped during training, reducing the network's dependence on the training data. A lower dropout value does not affect the model's training

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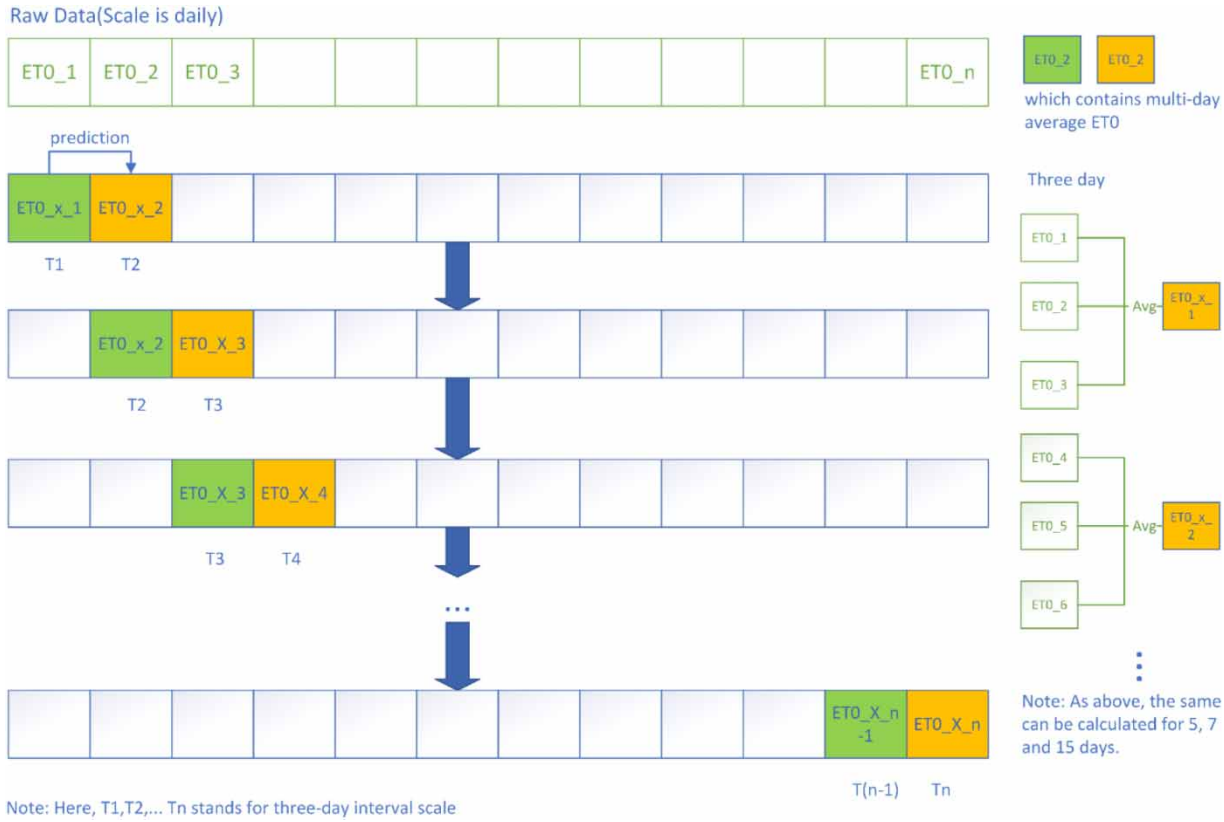

Figure 5 | ET_0 periodic forecast structure.

stability while helping to prevent overfitting and improve generalization. This study used the Adam optimizer for all DL models. Adam has shown superior performance in model training, as validated by Kingma & Ba (2014). In this study, the forecast is a scalar value, resulting in a single prediction output for the task, so Dense is set to 1. The mean squared error (MSE) was employed as the loss function to monitor the error between the actual values continuously and forecasted values in real-time, aiding in the tuning of model parameters (Kingma & Ba 2014). The batch_size represents the sample size used in each training iteration, while epochs denote the number of iterations for model training. The epochs and batch_size values are the optimal model parameters determined by GS. Suitable epochs and batch_size can enhance the model's prediction speed and stability. The activation function for all DL models is set as a rectified linear unit (ReLU), a nonlinear function with a range from 0 to 1. This function aids the model in learning more complex data relationships. Specific parameters are detailed in Table 2.

2.7. Evaluation of model performance

The accuracy and performance of the research models in forecasting different periods of ET_0 were determined using statistical metrics. We used four commonly employed evaluation metrics, along with a combined accuracy (CA) metric that better reflects the stability of the model's predictions, namely, R^2, mean absolute error (MAE),RMSE, Nash–Sutcliffe efficiency (NSE), and CA (Adnan *et al.* 2019), were employed for the evaluation and comparison of the results. Additionally, we used predicted absolute error (PAE) to assess the predictive performance of the model. The specific mathematical expressions are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_{i,\text{obs}} - y_{i,\text{pre}})^2}{\sum_{i=1}^{n} (y_{i,\text{obs}} - \overline{y_{i,\text{o}}})^2} \tag{21}$$

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Table 2 | Specific model parameters for ET_0 forecasting at the Nanchang, Xinjian, and Dongxiang stations

Station name	Models	Parameters
Nanchang, Xinjian, Dongxiang	XGB	■ Objective = reg:squarederror ■ n_estimators = 42
	SVM	■ kernel = RBF
	RF	■ n_estimators = 100, oob_score = True ■ n_jobs = 1, random_state = 50 ■ max_features = 1.0, min_samples_leaf = 10
	CNN	filters = 32, kernel_size = 1, activation = ReLU pool_size = 1, Dropout = 0.1 optimizer = Adam loss = mse, epochs = 40
	LSTM	units = 150, activation = ReLU, Dropout = 0.1 Dense = 1, optimizer = Adam, loss = mse epochs = 40, batch size = 32
	Bi-LSTM	units = 150, activation = ReLU, Dropout = 0.1 Dense = 1, optimizer = Adam, loss = mse epochs = 30, batch_size = 64
	CNN-LSTM	filters = 256, kernel_size = 1, activation = ReLU Dropout = 0.1, Dense = 1 LSTM (units = 150, activation = ReLU) Dense = 1, optimizer = Adam, loss = mse epochs = 30, batch_size = 32
	CNN-Bi-LSTM	filters = 256, kernel_size = 1, activation = ReLU Dropout = 0.1, Dense = 1 Bidirectional (LSTM (units = 150, activation = ReLU) Dense = 1, optimizer = Adam, loss = mse epochs = 40, batch_size = 64

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_{i,\mathrm{obs}} - y_{i,\mathrm{pre}}| \tag{22}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_{i,\mathrm{obs}} - y_{i,\mathrm{pre}})^2} \tag{23}$$

$$\mathrm{NSE} = 1 - \frac{\sum_{i=1}^{n} (y_{i,\mathrm{obs}} - y_{i,\mathrm{pre}})^2}{\sum_{i=1}^{n} (y_{i,\mathrm{obs}} - \overline{y_{i,\mathrm{o}}})^2} \tag{24}$$

$$\mathrm{CA} = 0.33(\mathrm{RMSE} + \mathrm{MAE} + (1 - R^2)) \tag{25}$$

$$\mathrm{PAE} = |y_{i,\mathrm{obs}} - y_{i,\mathrm{pre}}| \tag{26}$$

where $y_{i,\mathrm{obs}}$, $y_{i,\mathrm{pre}}$, and $y_{i,\mathrm{o}}$ represent the observed, forecasted, and mean values of ET_0, respectively. The range of R^2 is from 0 to 1, with a higher value indicating higher forecasting accuracy. On the contrary, MAE and RMSE have a range from 0 to positive infinity. When the forecasted value is close to the actual value, the numerator gradually approaches 0. In contrast, a larger numerator indicates a greater difference between the forecasted and actual values, leading to larger results. Lower MAE and RMSE values represent better forecasting accuracy and performance of the model. NSE and R^2 share the same formulaic structure, but the range of NSE is $(-\infty, 1]$. An NSE value of 1 indicates a perfectly accurate model prediction, while lower values signify poorer predictive performance. The CA value ranges from $(0, +\infty]$. A smaller CA value indicates greater stability in the model's forecasting performance. Figure 6 shows a flowchart of the model used in this study.

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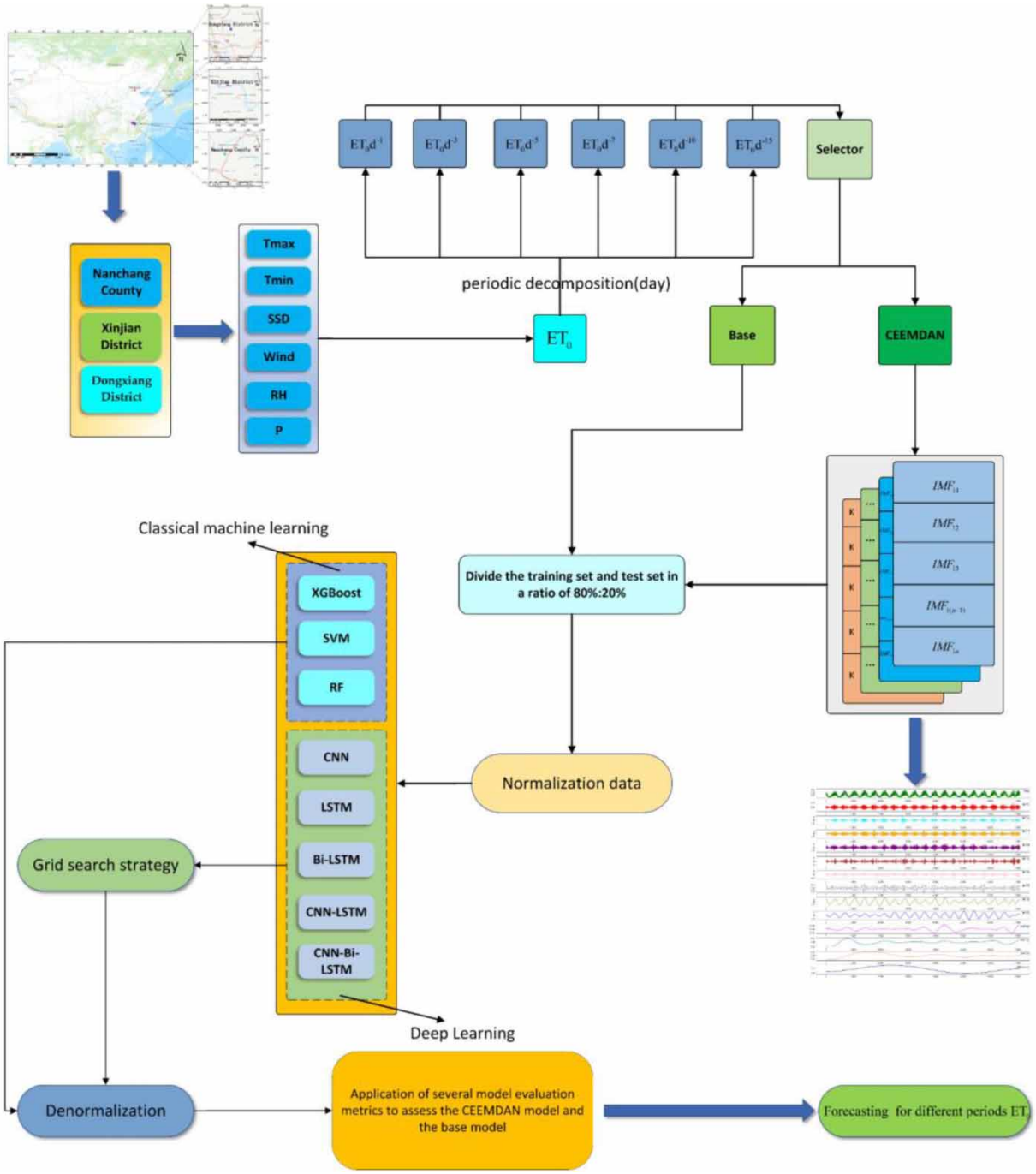

Figure 6 | Model flowchart for forecasting ET_0. Note that replacing CEEMDAN in the figure with EEMD and CEEMD results in the corresponding forecast structures.

3. RESULTS

3.1. Comparing the forecast accuracy of various models at different periods

To validate the importance of EEMD, CEEMD, and CEEMDAN decomposition algorithms for ET_0 prediction, an initial comparison was conducted to evaluate the predictive accuracy of various models without using the CEEMDAN decomposition algorithm over different time periods. Tables 3–5 describe the statistical results of forecasting ET_0 using three conventional ML models and five DL models in Nanchang, Dongxiang, and Xinjian, respectively. These tables present the

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Table 3 | The statistical values of ET_0 forecasting for different periods by various models in Nanchang station

Model	Statistical Indicators					Statistical Indicators				
ET_0 Period	R^2	MAE $mm\ d^-$	$RMSE$ $mm\ d^-$	NSE	CA	R^2	MAE $mm\ 3d$	$RMSE$ $mm\ 3d$	NSE	CA
	1-day					3-day				
XGB	0.9565	0.2718	0.3604	0.9565	0.2230	0.8851	0.3786	0.5462	0.8851	0.3431
SVM	**0.9593**	**0.3242**	**0.3488**	**0.9593**	**0.2355**	**0.9461**	**0.3328**	**0.3742**	**0.9461**	**0.2511**
RF	0.9509	0.2895	0.3830	0.9509	0.2381	0.9255	0.3448	0.4398	0.9255	0.2835
CNN	0.9726	0.2291	0.2860	0.9726	0.1790	0.9525	0.2790	0.3512	0.9525	0.2236
LSTM	0.9749	0.2268	0.2739	0.9749	0.1735	0.9668	0.2583	0.2938	0.9668	0.1931
Bi-LSTM	0.9768	0.2016	0.2630	0.9768	0.1610	0.9607	0.2515	0.3194	0.9607	0.2014
CNN-LSTM	0.9753	0.2422	0.2717	0.9753	0.1777	0.9425	0.2930	0.3863	0.9425	0.2431
CNN-Bi-LSTM	0.9763	0.2428	0.2661	0.9763	0.1758	0.9537	0.2915	0.3468	0.9537	0.2259
	5-day unit $mm\ 5d^{-1}$					7-day unit $mm\ 7d^{-1}$				
XGB	0.8990	0.3529	0.491	0.8990	0.3118	0.9156	0.3284	0.4407	0.9156	0.2817
SVM	**0.9447**	**0.2961**	**0.3634**	**0.9447**	**0.2359**	**0.9482**	**0.2626**	**0.3451**	**0.9482**	**0.2176**
RF	0.9352	0.3003	0.3934	0.9352	0.2503	0.9288	0.3238	0.4047	0.9288	0.2639
CNN	0.9635	0.2302	0.2952	0.9635	0.1854	0.9556	0.2438	0.3197	0.9556	0.2006
LSTM	0.9698	0.2245	0.2686	0.9698	0.1727	0.9669	0.2356	0.2761	0.9669	0.1798
Bi-LSTM	0.9650	0.2314	0.2888	0.9650	0.1832	0.9662	0.2326	0.279	0.9662	0.1800
CNN-LSTM	0.9454	0.2401	0.3611	0.9454	0.2164	0.9591	0.2356	0.3069	0.9591	0.1925
CNN-Bi-LSTM	0.9605	0.2321	0.307	0.9605	0.1909	0.9549	0.2667	0.3222	0.9549	0.2092
	10-day unit $mm\ 10d^{-1}$					15-day unit $mm\ 15d^{-1}$				
XGB	0.8327	0.4352	0.6067	0.8327	0.3990	0.8599	0.3970	0.5409	0.8599	0.3557
SVM	**0.9503**	**0.2538**	**0.3306**	**0.9503**	**0.2093**	0.9491	0.2103	0.3259	0.9491	0.1937
RF	0.9271	0.3110	0.4006	0.9271	0.2589	**0.9512**	**0.2393**	**0.3192**	**0.9512**	**0.2004**
CNN	0.9651	0.2247	0.277	0.9651	0.1771	0.6545	0.7295	0.8493	0.6545	0.6350
LSTM	0.9685	0.2263	0.2634	0.9685	0.1720	0.9571	0.2593	0.2994	0.9571	0.1985
Bi-LSTM	0.9738	0.2035	0.24	0.9738	0.1550	0.9718	0.2102	0.2427	0.9718	0.1588
CNN-LSTM	0.9609	0.2112	0.2932	0.9609	0.1794	0.9657	0.1982	0.2676	0.9657	0.1650
CNN-Bi-LSTM	0.9476	0.2248	0.3396	0.9476	0.2035	0.9631	0.1770	0.2774	0.9631	0.1621

Note: In the table, the cells with a yellow background represent the best forecasting results, while the text in bold indicates the best forecasting results among the three conventional ML models.

statistical analysis of ET_0 forecasting for the XGBoost (XGB), SVM, RF, CNN, LSTM, Bi-LSTM, CNN-LSTM, and CNN-Bi-LSTM models during the testing phase under six different periods: 1, 3, 5, 7, 10, and 15 days.

As shown in Table 3, the forecast results vary significantly depending on ET_0 forecasting models and ET_0 periods. Compared with ET_0 forecasting for multiple periods, the forecasting for ET_0 with a period of 1 day achieved the highest R^2 (0.9768) and the lowest MAE, NSE, CA, and RMSE, which are 0.2016 mm day^{-1}, 0.2630 mm day^{-1}, 0.9768, and 0.161, respectively. The forecasting model that achieved the best results is Bi-LSTM. Compared with several conventional ML models, including XGB, SVM, and RF, the DL-based Bi-LSTM demonstrates higher R^2 and lower MAE and RMSE in forecasting different periods of ET_0. It is worth noting that, in the forecasting of ET_0 with a period of 10 days, Bi-LSTM's R^2 is higher than the model with the lowest forecasting accuracy, XGB, by 16.94%. In addition, the MAE, RMSE, NSE, and CA of Bi-LSTM are respectively 25.83, 37.04, 38.51, and 16.94% higher than those of XGB in terms of forecasting performance. Table 3 shows that, when forecasting ET_0 for periods of 3, 5, and 7 days, the best-performing model is LSTM. Comparing the results of Bi-LSTM and LSTM for ET_0 forecasting with periods of 3, 5, and 7 days, the difference in R^2 forecasting results

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Table 4 | The statistical values of ET_0 forecasting for different periods by various models in Dongxiang station

Model	Statistical Indicators					Statistical Indicators				
ET_0 Period	R^2	MAE $mm\ d^-$	$RMSE$ $mm\ d^-$	NSE	CA	R^2	MAE $mm\ 3d$	$RMSE$ $mm\ 3d$	NSE	CA
	1-day					3-day				
XGB	0.9458	0.3108	0.4281	0.9458	0.2617	0.8722	0.4181	0.6049	0.8722	0.2684
SVM	**0.9514**	**0.3755**	**0.4051**	**0.9514**	**0.2736**	**0.9298**	**0.3879**	**0.4483**	**0.9298**	**0.3101**
RF	0.9464	0.3312	0.4256	0.9464	0.2674	0.9070	0.3848	0.516	0.9070	0.2867
CNN	0.9631	0.2918	0.3533	0.9631	0.2251	0.9387	0.3281	0.4189	0.9387	0.3000
LSTM	0.9746	0.2400	0.2932	0.9746	0.1843	0.9474	0.3217	0.3881	0.9474	0.3081
Bi-LSTM	0.9770	0.2491	0.2786	0.9770	0.1817	0.9458	0.3151	0.3941	0.9458	0.3039
CNN-LSTM	0.9463	0.3301	0.4261	0.9463	0.2673	0.9398	0.3323	0.4152	0.9398	0.3026
CNN-Bi-LSTM	0.9612	0.3203	0.362	0.9612	0.2380	0.9439	0.3109	0.4009	0.9439	0.3003
	5-day unit $mm\ 5d^{-1}$					7-day unit $mm\ 7d^{-1}$				
XGB	0.8833	0.3678	0.5563	0.8833	0.3435	0.8517	0.4350	0.6069	0.8517	0.2733
SVM	**0.9350**	**0.3191**	**0.4151**	**0.9350**	**0.2637**	**0.9341**	**0.3162**	**0.4045**	**0.9341**	**0.3009**
RF	0.9226	0.3287	0.4531	0.9226	0.2835	0.9167	0.3452	0.4548	0.9167	0.2938
CNN	0.9482	0.2733	0.3707	0.9482	0.2296	0.9381	0.3186	0.3921	0.9381	0.3057
LSTM	0.9506	0.3000	0.3619	0.9506	0.2347	0.9545	0.2905	0.3361	0.9545	0.3150
Bi-LSTM	0.9517	0.2841	0.3578	0.9517	0.2278	0.9534	0.2862	0.3402	0.9534	0.3122
CNN-LSTM	0.9161	0.2822	0.4719	0.9161	0.2765	0.9383	0.2644	0.3915	0.9383	0.2881
CNN-Bi-LSTM	0.9264	0.2699	0.442	0.9264	0.2592	0.9318	0.2729	0.4115	0.9318	0.2843
	10-day unit $mm\ 10d^{-1}$					15-day unit $mm\ 15d^{-1}$				
XGB	0.8351	0.4378	0.6237	0.8351	0.4047	0.8287	0.4544	0.6207	0.8287	0.2751
SVM	**0.9451**	**0.2323**	**0.3599**	**0.9451**	**0.2135**	**0.9316**	**0.2477**	**0.3922**	**0.9316**	**0.2823**
RF	0.9323	0.2999	0.3995	0.9323	0.2531	0.9186	0.2866	0.4279	0.9186	0.2834
CNN	0.9471	0.2804	0.3534	0.9471	0.2266	0.7709	0.5993	0.7178	0.7709	0.2909
LSTM	0.9673	0.2408	0.2779	0.9673	0.1820	0.9539	0.2793	0.3219	0.9539	0.3159
Bi-LSTM	0.9672	0.2391	0.2783	0.9672	0.1816	0.9677	0.2320	0.2696	0.9677	0.3176
CNN-LSTM	0.9589	0.2308	0.3115	0.9589	0.1925	0.9135	0.2605	0.441	0.9135	0.2704
CNN-Bi-LSTM	0.9449	0.2205	0.3607	0.9449	0.2100	0.8925	0.3216	0.4917	0.8925	0.2739

Note: In the table, the cells with a yellow background represent the best forecasting results, while the text in bold indicates the best forecasting results among the three conventional ML models.

is at the level of the third decimal place, indicating a difference of 0.001. Due to the varying patterns of ET_0 at different periods, the optimal forecasting model does not achieve uniformity. However, with the statistical difference at the 0.001 level, it can be considered an approximate forecasting accuracy in both statistical and practical terms. From Table 5, we can see that when forecasting ET_0 for different periods, the prediction accuracy of DL algorithms, precisely the LSTM and Bi-LSTM models, is higher than that of ML models such as XGB, SVM, and RF. This indicates that DL algorithms outperform ML algorithms in the forecasting of ET_0 based on time series. In Table 3, the highest forecasting accuracy is achieved by Bi-LSTM for forecasting ET_0 for 1 day. When grouped by forecasting period and sorted in descending order of forecasting accuracy, the sequence is as follows: Bi-LSTM forecasting ET_0 with a period of 1 day, Bi-LSTM forecasting ET_0 with a period of 10 days, Bi-LSTM forecasting ET_0 with a period of 15 days, LSTM forecasting ET_0 with a period of 5 days, LSTM forecasting ET_0 with a period of 7 days, and LSTM forecasting ET_0 with a period of 3 days. Tables 4 and 5 show similar trends in forecasting accuracy as Table 3. Table 3 shows that Bi-LSTM has an advantage in forecasting short periods (1 day) of ET_0 and longer periods (10 and 15 days) of ET_0 compared with other models. On the other hand, LSTM performs better in

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Table 5 | The statistical values of ET_0 forecasting for different periods by various models in Xinjian station

Model	Statistical Indicators					Statistical Indicators				
ET_0 Period	R^2	MAE $mm\ d^-$	$RMSE$ $mm\ d^-$	NSE	CA	R^2	MAE $mm\ 3d$	$RMSE$ $mm\ 3d$	NSE	CA
	1-day					3-day				
XGB	0.9402	0.2836	0.4077	0.9402	0.2479	0.9123	0.3218	0.4619	0.9123	0.2876
SVM	**0.9601**	**0.3140**	**0.3331**	**0.9601**	**0.2267**	**0.9455**	**0.3119**	**0.3641**	**0.9455**	**0.2411**
RF	0.9500	0.2848	0.3729	0.9500	0.2335	0.9274	0.3190	0.4202	0.9274	0.2679
CNN	0.9687	0.2324	0.295	0.9687	0.1844	0.9578	0.2635	0.3204	0.9578	0.2066
LSTM	0.9760	0.2241	0.2584	0.9760	0.1671	0.9668	0.2583	0.2938	0.9668	0.1931
Bi-LSTM	0.9740	0.2072	0.2686	0.9740	0.1656	0.9623	0.2571	0.3028	0.9623	0.1972
CNN-LSTM	0.9645	0.2625	0.3141	0.9645	0.2020	0.9512	0.2878	0.3446	0.9512	0.2248
CNN-Bi-LSTM	0.9535	0.2862	0.3595	0.9535	0.2284	0.9297	0.2669	0.4137	0.9297	0.2478
	5-day unit $mm\ 5d^{-1}$					7-day unit $mm\ 7d^{-1}$				
XGB	0.8526	0.3802	0.5715	0.8526	0.3627	0.8315	0.3883	0.6042	0.8315	0.3831
SVM	**0.9222**	**0.3156**	**0.4153**	**0.9222**	**0.2669**	**0.9124**	**0.3069**	**0.4357**	**0.9124**	**0.2740**
RF	0.9093	0.3105	0.4485	0.9093	0.2804	0.8918	0.3304	0.4841	0.8918	0.3045
CNN	0.9550	0.2447	0.316	0.9550	0.1999	0.9647	0.2341	0.2767	0.9647	0.1802
LSTM	0.9698	0.2245	0.2686	0.9698	0.1727	0.9669	0.2356	0.2761	0.9669	0.1798
Bi-LSTM	0.9559	0.2486	0.3125	0.9559	0.1997	0.9623	0.2339	0.2859	0.9623	0.1840
CNN-LSTM	0.9000	0.2857	0.4709	0.9000	0.2827	0.9287	0.2630	0.3929	0.9287	0.2400
CNN-Bi-LSTM	0.9050	0.2784	0.4589	0.9050	0.2747	0.8911	0.2796	0.4857	0.8911	0.2885
	10-day unit $mm\ 10d^{-1}$					15-day unit $mm\ 15d^{-1}$				
XGB	0.8493	0.3783	0.5577	0.8493	0.3586	0.7905	0.4832	0.6438	0.7905	0.4410
SVM	**0.9292**	**0.2955**	**0.3822**	**0.9292**	**0.2470**	**0.9279**	**0.2998**	**0.3777**	**0.9279**	**0.2474**
RF	0.9236	0.3055	0.397	0.9236	0.2570	0.9185	0.2863	0.4016	0.9185	0.2539
CNN	0.9566	0.2226	0.2992	0.9566	0.1865	0.9209	0.3181	0.3957	0.9209	0.2617
LSTM	0.9639	0.2341	0.2731	0.9639	0.1793	0.9578	0.2635	0.3204	0.9578	0.2066
Bi-LSTM	0.9712	0.2054	0.2439	0.9712	0.1578	0.9702	0.2090	0.2427	0.9702	0.1589
CNN-LSTM	0.9318	0.2427	0.375	0.9318	0.2263	0.8965	0.2813	0.4525	0.8965	0.2763
CNN-Bi-LSTM	0.9342	0.2403	0.3685	0.9342	0.2226	0.9453	0.2203	0.329	0.9453	0.1993

Note: In the table, the cells with a yellow background represent the best forecasting results, while the text in bold indicates the best forecasting results among the three conventional ML models.

forecasting medium periods (3, 5, and 7 days) of ET_0 compared with other models. This pattern is consistent in both Tables 4 and 5. From the experimental results, Bi-LSTM demonstrates a more stable advantage in forecasting short and long periods of ET_0, while LSTM exhibits an advantage in forecasting medium periods of ET_0.

Tables 3–5 show that the forecast results follow similar patterns in different regions. For a detailed analysis based on Nanchang, scatter plots of different ET_0 values for various periods calculated by the FAO-56 PM model and forecasted by three ML and five DL models are shown in Figure 7. The figure shows that the studied models exhibit significant differences in forecasting ET_0 for different periods. In forecasting ET_0 for various periods, the values forecasted by the CNN, LSTM, Bi-LSTM, CNN-LSTM, and CNN-Bi-LSTM models are closer to those calculated by the FAO-56 PM equation. This indicates that models based on DL are more accurate in forecasting ET_0, showing a strong correlation with the values obtained from the FAO-56 PM equation. The scatter plot in Figure 7 provides a clearer view of the forecasting accuracy of different models for ET_0. In the forecasting of ET_0 for various periods, the dispersion of data points for XGB, SVM, and RF is significantly higher than that of other models in Figure 7. It can be observed in Figure 7 that when forecasting ET_0 for a 1-day period,

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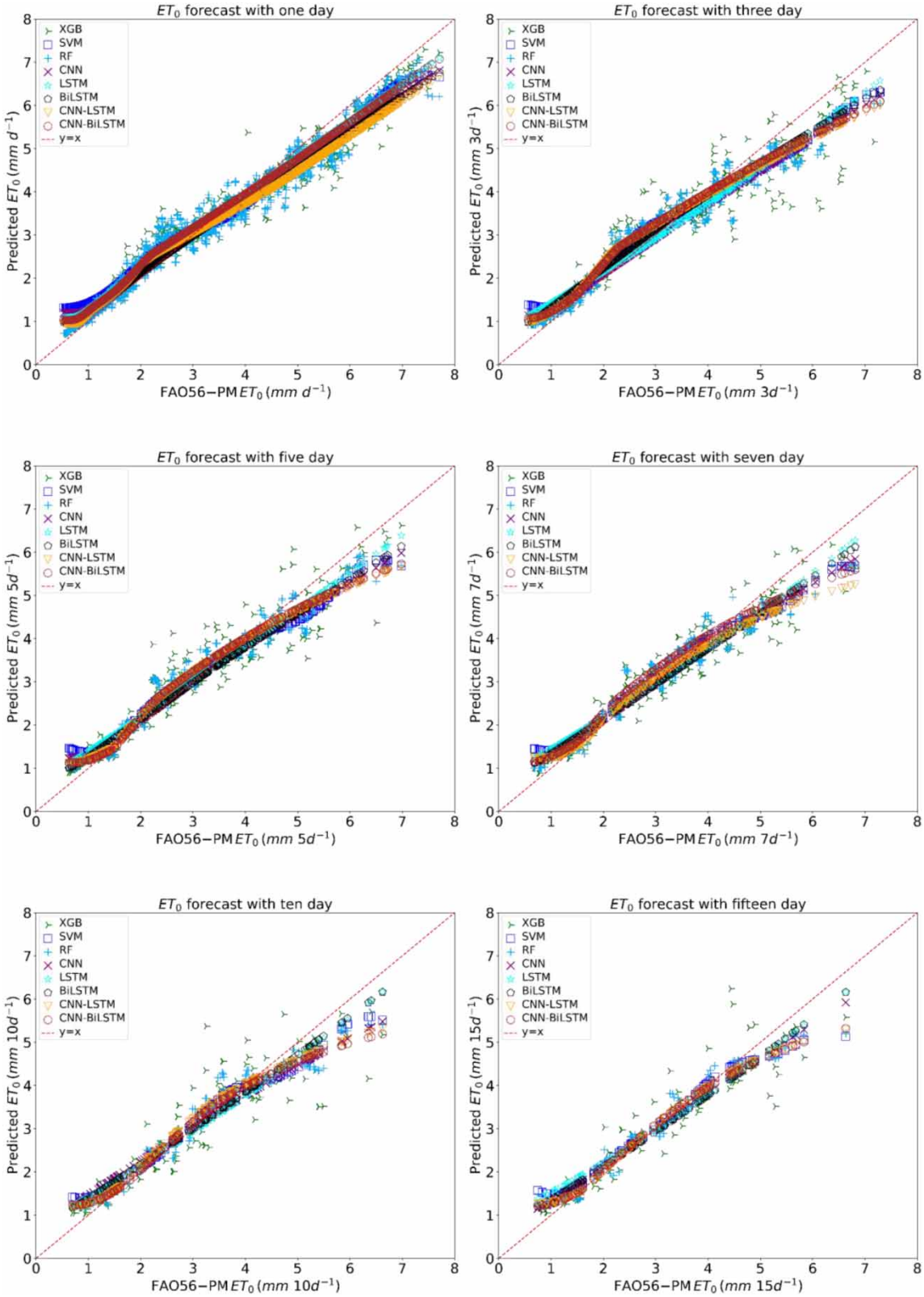

Figure 7 | Scatter plots of ET_0 forecasting for different periods by different models at Nanchang station.

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the dispersion of data points for each model is lower. This pattern is further evident in Figure 9. Additionally, in Figure 7, it is clear that for the forecasting of ET_0 over 3-, 5-, and 7-day periods, the forecasting trend of LSTM is closer to $y = x$, indicating that its forecasted values are closer to the actual values. However, in the forecasting of ET_0 over a 15-day period, Bi-LSTM is even closer to the $y = x$ line. This explains that the performance of models varies for different periods. It also indicates that LSTM and Bi-LSTM perform well forecasting ET_0 for different periods.

To further illustrate its periodicity and the forecasting relationship of LSTM and Bi-LSTM models, Figure 8 provides an analysis of the forecasting results (three-dimensional curve graph) of ET_0 for different periods in Nanchang, Dongxiang, and Xinjian.

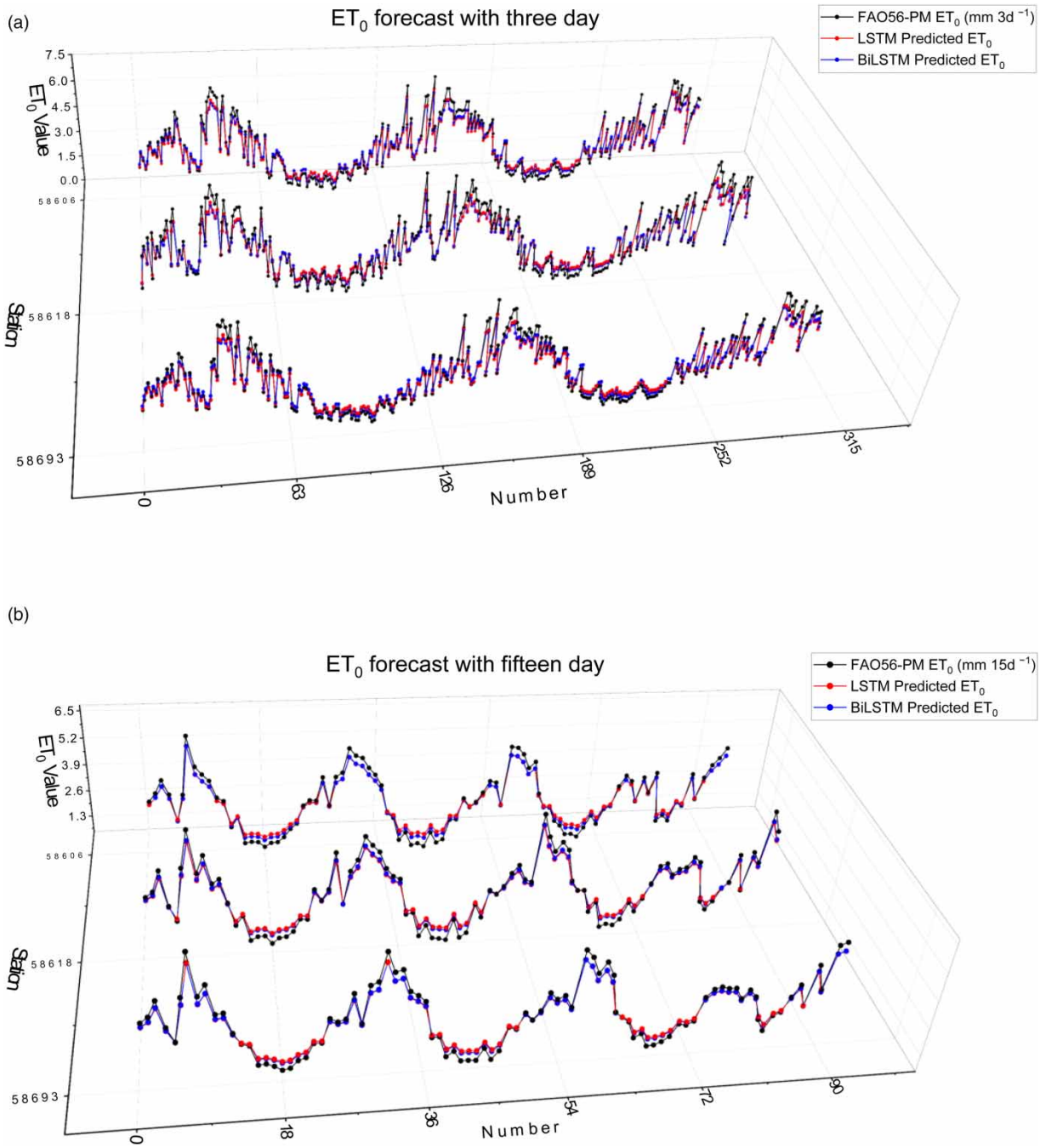

Figure 8 | Three-dimensional curve plots illustrate the forecasting of ET_0 for 3 (a) and 15 (b) days by LSTM and Bi-LSTM models at Nanchang, Dongxiang, and Xinjian stations.

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It is clear from the graph that for different stations, the two models show significant differences in forecasting ET_0 for the same period. From Figure 8, it is evident that for forecasting a period of 3 days of ET_0, when compared with the curve of the actual values, the forecasting curve of LSTM is closer to the actual values than Bi-LSTM for all three different stations. Conversely, from the graph of the forecasting period of 15 days for ET_0, it is clear that the forecasting curve of Bi-LSTM is closer to the actual curve of ET_0. Figure 8 effectively illustrates that Bi-LSTM and LSTM have different advantages in forecasting ET_0 for different periods. Although the difference in their forecasting accuracy is small, Figure 8 suggests that Bi-LSTM is more suitable for forecasting long-period ET_0. At the same time, LSTM is more suitable for forecasting shorter periods of ET_0. In this study, LSTM and Bi-LSTM accurately forecasted ET_0 over different periods. Short-term forecasts can provide more precise guidance for daily irrigation, while long-term forecasts contribute to long-term water resource planning.

3.2. Comparing the forecasting accuracy of CEEMDAN with the base models at different periods

This study aims to investigate whether adding the CEEMDAN algorithm on top of the base models can lead to higher forecasting accuracy and, consequently, to analyze the importance of CEEMDAN in forecasting ET_0. Tables 6–8 provide a clear overview of the forecasting accuracy of ET_0 for different stations and periods based on the CEEMDAN algorithm. Analysis of Table 6 reveals that CEEMDAN-Bi-LSTM achieved the best forecasting results when forecasting ET_0 for a period of 15 days. However, there is no consistent optimal forecasting model for different ET_0 periods. A different pattern from Tables 3–5 is observed, wherein the CEEMDAN-based algorithms, forecasting ET_0 for a 1-day period yielded better accuracy with CEEMDAN-XGB compared with the forecasting accuracy of CEEMDAN-CNN-LSTM, a DL-based model. The same pattern is observed when forecasting the Dongxiang and Xinjian stations. Tables 3–8 clearly show that in forecasting ET_0 for different periods, all DL-based algorithms outperform ML-based XGB, SVM, and RF. However, forecasting the ET_0 for a period of 1 day shows different results, which is in stark contrast to Tables 3–5. Figure 9 shows the distribution of the optimal models for forecasting for ET_0 different stations and periods between the base models and those based on CEEMDAN. Analysis of Figure 9 reveals that adding the CEEMDAN algorithm to the forecasting for different periods does not follow a consistent pattern, significantly reducing the stability of the forecasting and the universality of the models.

Through the analysis of Tables 3 and 6, in the forecasting of ET_0 with a period of 1 day in Nanchang, the best model of the base model, Bi-LSTM ($R^2 = 0.9768$, MAE $= 0.2016$ mm day^{-1}, RMSE $= 0.2630$ mm day^{-1}, NSE $= 0.9768$, CA $= 0.1610$), outperforms the best model based on CEEMDAN, CEEMDAN-CNN-Bi-LSTM ($R^2 = 0.8339$, MAE $= 0.5206$ mm day^{-1}, RMSE $=$ 0.7044 mm day^{-1}, NSE $= 0.8339$, CA $= 0.4591$), by 17.13% in R^2, and is lower in terms of MAE, RMSE, NSE, and CA by 158.23, 167.83, 17.13, and 24.74%, respectively. Tables 6–8 exhibit similar patterns. Based on Tables 3 and 6, a detailed analysis was conducted. Absolute forecast errors for ET_0 at Nanchang station with periods of 1 and 15 days were calculated. Box plots were used to visually analyze and compare the absolute forecast errors between the base and CEEMDAN models. We can clearly see from Figure 10 that the base model has fewer outliers in the absolute forecast errors, and, at the same time, the overall values of absolute forecast errors are smaller compared with the CEEMDAN models. This indicates that the base model has higher forecasting accuracy than various CEEMDAN models in forecasting ET_0 for different periods. To further analyze the forecasting stability between the base model and CEEMDAN, a statistical analysis of the distribution of the best forecast models for all stations and periods of ET_0 was conducted. From the histogram in Figure 9, it is clear that, compared with the base model, CEEMDAN models exhibit a highly uneven distribution of the best models in forecasting. This indicates that the CEEMDAN models for ET_0 forecasting are unstable, and the addition of CEEMDAN has yet to improve the generalization of the models. Instead, it has increased the inconsistency of the forecasting models, leading to a decrease in forecast stability.

3.3. Comparison of the forecasting performance of different decomposition models: EEMD, CEEMD, and CEEMDAN

To further validate the impact and significance of the CEEMDAN decomposition model on multi-period ET_0 forecasting in humid regions, this study introduced EEMD and CEEMD methods combined with the same ML models as CEEMDAN. The forecasting performance of EEMD and CEEMD-based hybrid ML models was analyzed for multi-period ET_0 predictions at the Nanchang, Dongxiang, and Xinjian stations. Experimental analysis revealed that EEMD and CEEMD demonstrated similar forecasting patterns and comparable performance across all three stations for multi-period ET_0 predictions. Therefore, detailed analyses of EEMD and CEEMD forecasting results were conducted based on the results from the Nanchang station. Table 9 presents the model evaluation metrics for the EEMD-based hybrid ML models at Nanchang station over six different periods. Except for ET_0 predictions at 1- and 3-day periods, EMMD-LSTM and EEMD-Bi-LSTM were identified as the best forecasting models for all other

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Table 6 | Statistical values of the CEEMDAN model for forecasting ET_0 over different periods at the Nanchang station

Model	Statistical Indicators					Statistical Indicators				
ET_0 Period	R^2	MAE $mm\ d^-$	$RMSE$ $mm\ d^-$	NSE	CA	R^2	MAE $mm\ 3d$	$RMSE$ $mm\ 3d$	NSE	CA
	1-day					3-day				
CEEMDAN-XGB	**0.8113**	**0.5399**	**0.7508**	**0.8113**	**0.4882**	**0.7838**	**0.5560**	**0.7494**	**0.7838**	**0.5021**
CEEMDAN-SVM	0.8072	0.5834	0.7589	0.8072	0.5066	0.7522	0.6324	0.8022	0.7522	0.5552
CEEMDAN-RF	0.7922	0.5795	0.7879	0.7922	0.5198	0.7712	0.5830	0.7708	0.7712	0.5223
CEEMDAN-CNN	0.8267	0.5360	0.7196	0.8267	0.4715	0.7836	0.5742	0.7496	0.7836	0.5083
CEEMDAN-LSTM	0.8266	0.5314	0.7196	0.8266	0.4701	0.7969	0.5465	0.7263	0.7969	0.4870
CEEMDAN-Bi-LSTM	0.8273	0.5330	0.7182	0.8273	0.4699	0.7958	0.5497	0.7282	0.7958	0.4891
CEEMDAN-CNN-LSTM	0.8027	0.5735	0.7677	0.8027	0.5077	0.7989	0.5539	0.7227	0.7989	0.4876
CEEMDAN-CNN-Bi-LSTM	0.8339	0.5206	0.7044	0.8339	0.4591	0.7900	0.5618	0.7386	0.7900	0.4984
	5-day unit $mm\ 5d^{-1}$					7-day unit $mm\ 7d^{-1}$				
CEEMDAN-XGB	0.7594	0.5853	0.7578	0.7594	0.5226	**0.7190**	**0.6139**	**0.8041**	**0.7190**	**0.5607**
CEEMDAN-SVM	0.7394	0.6330	0.7887	0.7394	0.5552	0.6946	0.6734	0.8382	0.6946	0.5996
CEEMDAN-RF	**0.7604**	**0.5858**	**0.7562**	**0.7604**	**0.5219**	0.7099	0.6285	0.8169	0.7099	0.5727
CEEMDAN-CNN	0.8146	0.5241	0.6653	0.8146	0.4537	0.7970	0.5420	0.6834	0.7970	0.4714
CEEMDAN-LSTM	0.8026	0.5387	0.6864	0.8026	0.4694	0.7565	0.5946	0.7484	0.7565	0.5235
CEEMDAN-Bi-LSTM	0.7969	0.5509	0.6963	0.7969	0.4786	0.7303	0.6279	0.7878	0.7303	0.5562
CEEMDAN-CNN-LSTM	0.7936	0.5587	0.7018	0.7936	0.4841	0.7215	0.6328	0.8004	0.7215	0.5649
CEEMDAN-CNN-Bi-LSTM	0.7851	0.5779	0.7162	0.7851	0.4980	0.7310	0.6252	0.7867	0.7310	0.5547
	10-day unit $mm\ 10d^{-1}$					15-day unit $mm\ 15d^{-1}$				
CEEMDAN-XGB	**0.7097**	**0.5995**	**0.7993**	**0.7097**	**0.5574**	**0.7376**	**0.5786**	**0.7402**	**0.7376**	**0.5218**
CEEMDAN-SVM	0.7051	0.6366	0.8056	0.7051	0.5732	0.7240	0.6071	0.7592	0.7240	0.5420
CEEMDAN-RF	0.7086	0.6117	0.8008	0.7086	0.5623	0.7121	0.6059	0.7753	0.7121	0.5508
CEEMDAN-CNN	0.7454	0.5949	0.7486	0.7454	0.5274	0.7357	0.6380	0.7429	0.7357	0.5429
CEEMDAN-LSTM	0.8520	0.4440	0.5707	0.8520	0.3837	0.8752	0.4147	0.5105	0.8752	0.3465
CEEMDAN-Bi-LSTM	0.8220	0.4884	0.6259	0.8220	0.4265	0.8872	0.3863	0.4853	0.8872	0.3249
CEEMDAN-CNN-LSTM	0.7654	0.5642	0.7185	0.7654	0.5007	0.7738	0.5455	0.6873	0.7738	0.4815
CEEMDAN-CNN-Bi-LSTM	0.7581	0.5712	0.7296	0.7581	0.5091	0.7698	0.5480	0.6933	0.7698	0.4856

Note: In the table, the cells with a yellow background represent the best forecasting results, while the text in bold indicates the best forecasting results among the three conventional ML models.

periods, consistent with the results observed for standalone models and CEEMDAN hybrid models. Moreover, Table 9 shows that the EEMD hybrid DL models outperformed the EEMD hybrid traditional ML models in forecasting accuracy.

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Table 7 | Statistical values of the CEEMDAN model for forecasting ET_0 over different periods at the Dongxiang station

Model	Statistical Indicators					Statistical Indicators				
ET_0 Period	R^2	MAE $mm\ d^-$	$RMSE$ $mm\ d^-$	NSE	CA	R^2	MAE $mm\ 3d$	$RMSE$ $mm\ 3d$	NSE	CA
	1-day					3-day				
CEEMDAN-XGB	**0.8129**	**0.5980**	**0.7952**	**0.8129**	**0.5215**	0.7171	0.6837	0.900	0.7171	0.6160
CEEMDAN-SVM	0.7927	0.6581	0.837	0.7927	0.5618	**0.7083**	**0.7286**	**0.914**	**0.7083**	**0.6383**
CEEMDAN-RF	0.7868	0.6414	0.8488	0.7868	0.5621	0.7186	0.7016	0.8976	0.7186	0.6206
CEEMDAN-CNN	0.8140	0.6056	0.7929	0.8140	0.5229	0.7615	0.6481	0.8265	0.7615	0.5653
CEEMDAN-LSTM	0.8176	0.5927	0.7851	0.8176	0.5149	0.7522	0.6616	0.8423	0.7522	0.5781
CEEMDAN-Bi-LSTM	0.8191	0.5939	0.7819	0.8191	0.5137	0.7542	0.6588	0.839	0.7542	0.5754
CEEMDAN-CNN-LSTM	0.8080	0.6134	0.8057	0.8080	0.5317	0.7641	0.6515	0.8219	0.7641	0.5641
CEEMDAN-CNN-Bi-LSTM	0.7941	0.6401	0.8342	0.7941	0.5545	0.7520	0.6645	0.8427	0.7520	0.5792
	5-day unit $mm\ 5d^{-1}$					7-day unit $mm\ 7d^{-1}$				
CEEMDAN-XGB	**0.7090**	**0.6856**	**0.8787**	**0.7090**	**0.6122**	**0.7096**	**0.6755**	**0.8492**	**0.7096**	**0.5990**
CEEMDAN-SVM	0.6698	0.7536	0.936	0.6698	0.6665	0.6896	0.7072	0.8779	0.6896	0.6255
CEEMDAN-RF	0.6868	0.7193	0.9115	0.6868	0.6415	0.6998	0.6990	0.8634	0.6998	0.6147
CEEMDAN-CNN	0.7540	0.6484	0.8079	0.7540	0.5618	0.8366	0.5112	0.637	0.8366	0.4328
CEEMDAN-LSTM	0.7340	0.6711	0.8400	0.7340	0.5864	0.7730	0.6119	0.7508	0.7730	0.5246
CEEMDAN-Bi-LSTM	0.7312	0.6750	0.8445	0.7312	0.5901	0.7612	0.6291	0.7701	0.7612	0.5405
CEEMDAN-CNN-LSTM	0.7384	0.6652	0.8331	0.7384	0.5808	0.7016	0.6901	0.8608	0.7016	0.6103
CEEMDAN-CNN-Bi-LSTM	0.7218	0.6910	0.8591	0.7218	0.6033	0.7348	0.6618	0.8115	0.7348	0.5737
	10-day unit $mm\ 10d^{-1}$					15-day unit $mm\ 15d^{-1}$				
CEEMDAN-XGB	0.6379	0.7055	0.9242	0.6379	0.6573	0.5727	0.7228	0.9803	0.5727	0.7030
CEEMDAN-SVM	**0.6945**	**0.6890**	**0.8490**	**0.6945**	**0.6084**	**0.6883**	**0.6743**	**0.8372**	**0.6883**	**0.6017**
CEEMDAN-RF	0.6673	0.6899	0.8859	0.6673	0.6298	0.6276	0.6953	0.9152	0.6276	0.6544
CEEMDAN-CNN	0.7707	0.5811	0.7355	0.7707	0.5101	0.8748	0.4048	0.5307	0.8748	0.3500
CEEMDAN-LSTM	0.8417	0.4857	0.611	0.8417	0.4142	0.8538	0.4563	0.5733	0.8538	0.3880
CEEMDAN-Bi-LSTM	0.8172	0.5242	0.6567	0.8172	0.4500	0.8665	0.4335	0.548	0.8665	0.3680
CEEMDAN-CNN-LSTM	0.7424	0.6227	0.7796	0.7424	0.5478	0.7072	0.6567	0.8115	0.7072	0.5811
CEEMDAN-CNN-Bi-LSTM	0.7357	0.6467	0.7896	0.7357	0.5612	0.6975	0.6562	0.8248	0.6975	0.5886

Note: In the table, the cells with a yellow background represent the best forecasting results, while the text in bold indicates the best forecasting results among the three conventional ML models.

Notably, for ET_0 predictions at 10- and 15-day forecasting periods, the EEMD-LSTM and EEMD-Bi-LSTM models outperformed EEMD-SVM in terms of R^2 by approximately 20 and 15.49%, respectively, and in terms of the comprehensive CA metric by approximately 39.47 and 34.37%, respectively. It is worth highlighting that EEMD demonstrated forecasting

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Table 8 | Statistical values of the CEEMDAN model for forecasting ET_0 over different periods at the Xinjian station

Model	Statistical Indicators					Statistical Indicators				
ET_0 Period	R^2	MAE $mm\ d^-$	$RMSE$ $mm\ d^-$	NSE	CA	R^2	MAE $mm\ 3d$	$RMSE$ $mm\ 3d$	NSE	CA
	1-day					3-day				
CEEMDAN-XGB	**0.8191**	**0.5194**	**0.7090**	**0.8191**	**0.4651**	**0.7678**	**0.5605**	**0.7517**	**0.7678**	**0.5097**
CEEMDAN-SVM	0.7949	0.5928	0.7550	0.7949	0.5125	0.7566	0.6029	0.7696	0.7566	0.5332
CEEMDAN-RF	0.8043	0.5471	0.7373	0.8043	0.4884	0.7624	0.5729	0.7605	0.7624	0.5184
CEEMDAN-CNN	0.8173	0.5406	0.7126	0.8173	0.4738	0.8049	0.5302	0.6891	0.8049	0.4668
CEEMDAN-LSTM	0.8295	0.5200	0.6882	0.8295	0.4550	0.8019	0.5258	0.6942	0.8019	0.4680
CEEMDAN-Bi-LSTM	0.8295	0.5215	0.6883	0.8295	0.4555	0.8007	0.5247	0.6964	0.8007	0.4687
CEEMDAN-CNN-LSTM	0.8118	0.5453	0.7232	0.8118	0.4807	0.8110	0.5204	0.6782	0.8110	0.4579
CEEMDAN-CNN-Bi-LSTM	0.8226	0.5303	0.7022	0.8226	0.4653	0.8038	0.5274	0.691	0.8038	0.4668
	5-day unit $mm\ 5d^{-1}$					7-day unit $mm\ 7d^{-1}$				
CEEMDAN-XGB	0.7410	0.5809	0.7577	0.7410	0.5272	0.7088	0.6032	0.7943	0.7088	0.5573
CEEMDAN-SVM	0.7366	0.6332	0.7641	0.7366	0.5480	**0.7155**	**0.6280**	**0.7851**	**0.7155**	**0.5602**
CEEMDAN-RF	**0.7516**	**0.5822**	**0.7420**	**0.7516**	**0.5190**	0.7042	0.6160	0.8006	0.7042	0.5651
CEEMDAN-CNN	0.7819	0.5687	0.6953	0.7819	0.4891	0.8202	0.4820	0.6241	0.8202	0.4243
CEEMDAN-LSTM	0.7838	0.5570	0.6922	0.7838	0.4836	0.7904	0.5357	0.6739	0.7904	0.4683
CEEMDAN-Bi-LSTM	0.7832	0.5578	0.6933	0.7832	0.4844	0.7801	0.5522	0.6902	0.7801	0.4826
CEEMDAN-CNN-LSTM	0.7815	0.5603	0.6960	0.7815	0.4867	0.7630	0.5749	0.7166	0.7630	0.5044
CEEMDAN-CNN-Bi-LSTM	0.7835	0.5585	0.6928	0.7835	0.4844	0.7677	0.5698	0.7095	0.7677	0.4988
	10-day unit $mm\ 10d^{-1}$					15-day unit $mm\ 15d^{-1}$				
CEEMDAN-XGB	0.7050	0.5841	0.7802	0.7050	0.5476	0.6957	0.5598	0.7759	0.6957	0.5412
CEEMDAN-SVM	0.7148	0.6107	0.7671	0.7148	0.5488	**0.7195**	**0.5787**	**0.7449**	**0.7195**	**0.5294**
CEEMDAN-RF	**0.7319**	**0.5679**	**0.7438**	**0.7319**	**0.5213**	0.7088	0.5570	0.759	0.7088	0.5304
CEEMDAN-CNN	0.8687	0.4065	0.5205	0.8687	0.3492	0.7146	0.6135	0.7513	0.7146	0.5446
CEEMDAN-LSTM	0.8383	0.4506	0.5777	0.8383	0.3927	0.8656	0.4080	0.5155	0.8656	0.3491
CEEMDAN-Bi-LSTM	0.8193	0.4778	0.6107	0.8193	0.4188	0.8793	0.3770	0.4887	0.8793	0.3255
CEEMDAN-CNN-LSTM	0.7735	0.5325	0.6836	0.7735	0.4761	0.7941	0.4960	0.6382	0.7941	0.4422
CEEMDAN-CNN-Bi-LSTM	0.7670	0.5415	0.6933	0.7670	0.4844	0.7830	0.5013	0.6552	0.7830	0.4533

Note: In the table, the cells with a yellow background represent the best forecasting results, while the text in bold indicates the best forecasting results among the three conventional ML models.

performance comparable to CEEMDAN across different forecasting periods for ET_0. Moreover, in certain forecasting periods, the EEMD method showed superior performance compared with CEEMDAN. However, significant differences remained when compared with the baseline model. For the 10-day forecasting period, Bi-LSTM emerged as the

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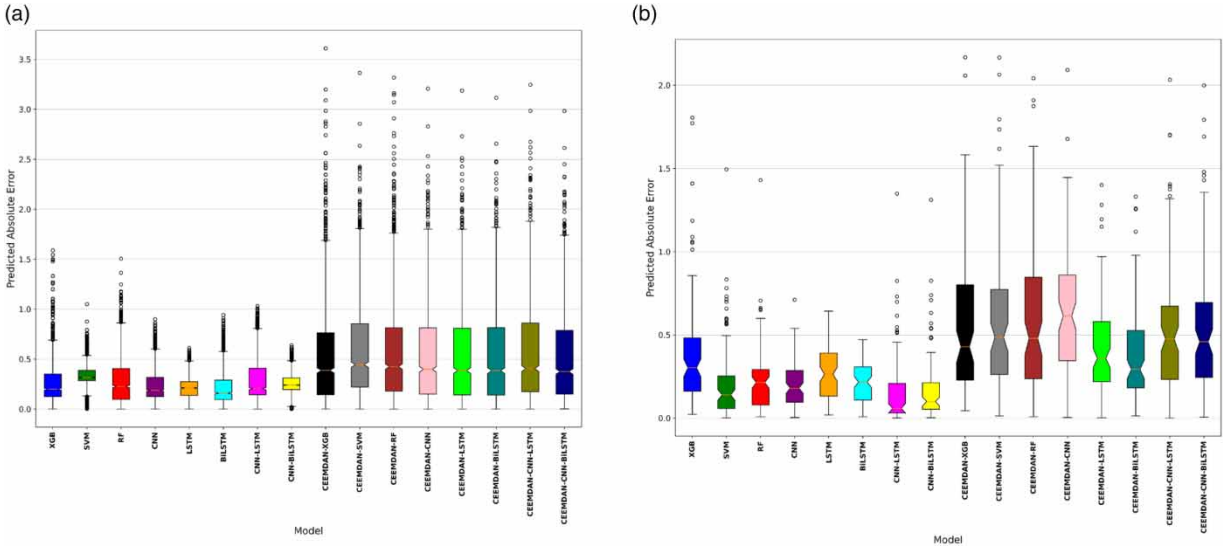

Figure 9 | Box plots of absolute forecast errors between the base model and the CEEMDAN model at Nanchang station for periods of 1 day (a) and 15 days (b).

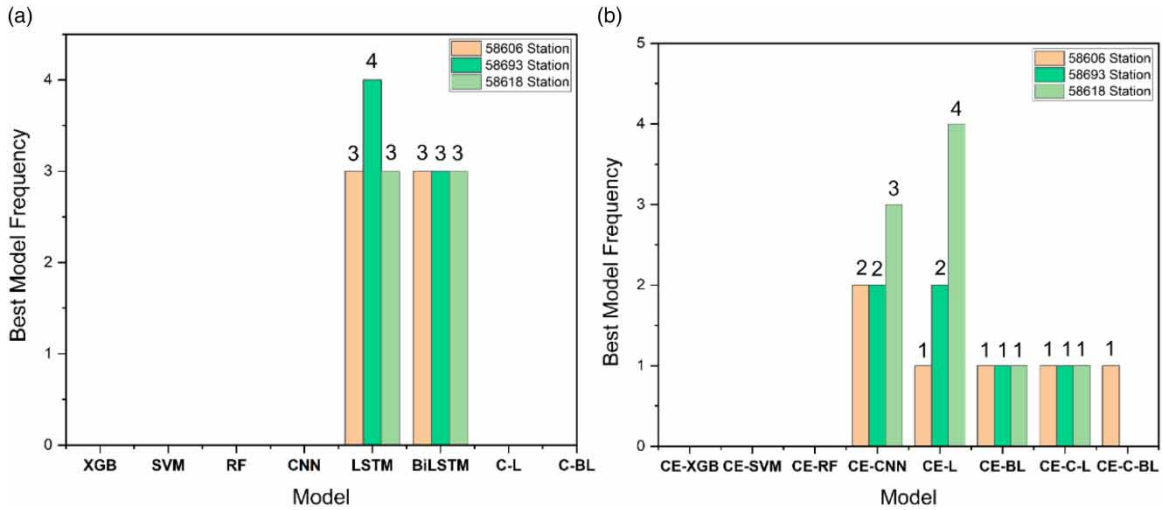

Figure 10 | Distribution of the best models for the forecasting of ET_0 for different cycles by the base model (a) and the CEEMDAN model (b) at Nanchang, Dongxiang, and Xinjian stations. *Note*: C-L: CNN-LSTM; C-BL: CNN-Bi-LSTM; CE-XGB: CEEMDAN-XGB; CE-L: CEEMDAN-LSTM; CE-C-L: CEEMDAN-CNN-LSTM.

best-performing model. Compared with EEMD-Bi-LSTM, Bi-LSTM achieved superior performance in the comprehensive CA metric, with its forecasting accuracy exceeding that of EEMD-Bi-LSTM by approximately 53.87% under the same forecasting conditions.

In addition to EEMD, this study employed CEEMD combined with the same models as CEEMDAN to analyze the effects of different decomposition methods on ET_0 predictions across various time periods. Table 10 presents the detailed evaluation metrics for the predictive performance of the CEEMD method. As shown in Table 10, CEEMD-LSTM emerged as the best-performing model for the 5-, 7-, and 10-day forecasting periods, accounting for three out of the six best models across all periods. This result aligns with the predictive patterns observed for the baseline model, the CEEMDAN-based hybrid models, and the EEMD-based hybrid models. LSTM and Bi-LSTM were consistently identified as the most effective models for multi-period ET_0 prediction in humid regions. Furthermore, hybrid DL models outperformed hybrid traditional

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Table 9 | Statistical values of the EEMD model for forecasting ET_0 over different periods at the Nanchang station

Model	Statistical Indicators					Statistical Indicators				
ET_0 Period	R^2	MAE $mm\ d^-$	$RMSE$ $mm\ d^-$	NSE	CA	R^2	MAE $mm\ 3d$	$RMSE$ $mm\ 3d$	NSE	CA
	1-day					3-day				
EEMD-XGB	**0.8187**	**0.5364**	**0.7359**	**0.8187**	**0.4797**	0.7255	0.6051	0.8452	0.7255	0.5692
EEMD-SVM	0.8076	0.5835	0.7580	0.8076	0.5062	**0.7440**	**0.6223**	**0.8162**	**0.7440**	**0.5592**
EEMD-RF	0.7875	0.5987	0.7966	0.7875	0.5306	0.7000	0.6529	0.8836	0.7000	0.6060
EEMD-CNN	0.8239	0.5355	0.7252	0.8239	0.4741	0.7674	0.5822	0.7780	0.7674	0.5256
EEMD-LSTM	0.8264	0.5329	0.7201	0.8264	0.4708	0.7671	0.5706	0.7785	0.7671	0.5220
EEMD-Bi-LSTM	0.8248	0.5322	0.7234	0.8248	0.4722	0.7620	0.5777	0.7870	0.7620	0.5289
EEMD-CNN-LSTM	0.8466	0.4964	0.6769	0.8466	0.4378	0.7628	0.5829	0.7858	0.7628	0.5300
EEMD-CNN-Bi-LSTM	0.8268	0.5272	0.7193	0.8268	0.4685	0.7540	0.5874	0.8001	0.7540	0.5390
	5-day unit $mm\ 5d^{-1}$					7-day unit $mm\ 7d^{-1}$				
EEMD-XGB	0.7441	0.5805	0.7810	0.7441	0.5337	**0.7433**	**0.5899**	**0.7685**	**0.7433**	**0.5330**
EEMD-SVM	**0.7539**	**0.5986**	**0.7659**	**0.7539**	**0.5315**	0.7325	0.6048	0.7845	0.7325	0.5467
EEMD-RF	0.7231	0.6120	0.8123	0.7231	0.5614	0.7231	0.6120	0.8123	0.7231	0.5614
EEMD-CNN	0.7648	0.5568	0.7486	0.7648	0.5084	0.7787	0.5587	0.7135	0.7787	0.4928
EEMD -LSTM	0.7838	0.5380	0.7178	0.7838	0.4858	0.7928	0.5279	0.6905	0.7928	0.4704
EEMD-Bi-LSTM	0.7725	0.5510	0.7362	0.7725	0.4999	0.7636	0.5647	0.7375	0.7636	0.5078
EEMD-CNN-LSTM	0.7697	0.5596	0.7409	0.7697	0.5052	0.7312	0.6078	0.7864	0.7312	0.5488
EEMD-CNN-Bi-LSTM	0.7187	0.6289	0.8187	0.7187	0.5706	0.7438	0.5940	0.7678	0.7438	0.5340
	10-day unit $mm\ 10d^{-1}$					15-day unit $mm\ 15d^{-1}$				
EEMD-XGB	0.6804	0.6212	0.8387	0.6804	0.5872	0.6771	0.6104	0.8173	0.6771	0.5777
EEMD-SVM	**0.7437**	**0.5947**	**0.7510**	**0.7437**	**0.5287**	**0.7711**	**0.5241**	**0.6881**	**0.7711**	**0.4756**
EEMD-RF	0.7043	0.6202	0.8067	0.7043	0.5684	0.7100	0.5895	0.7745	0.7100	0.5458
EEMD-CNN	0.8246	0.4919	0.6212	0.8246	0.4252	0.8635	0.4135	0.5314	0.8635	0.3569
EEMD-LSTM	0.8924	0.3754	0.4866	0.8924	0.3200	0.8879	0.3808	0.4816	0.8879	0.3216
EEMD-Bi-LSTM	0.8828	0.3929	0.5079	0.8828	0.3360	0.8905	0.3606	0.4759	0.8905	0.3122
EEMD-CNN-LSTM	0.7493	0.5809	0.7428	0.7493	0.5195	0.7834	0.5083	0.6694	0.7834	0.4601
EEMD-CNN-Bi-LSTM	0.7240	0.6126	0.7795	0.7240	0.5505	0.7855	0.5063	0.6660	0.7855	0.4576

Note: In the table, the cells with a yellow background represent the best forecasting results, while the text in bold indicates the best forecasting results among the three conventional ML models.

ML models, a trend consistent with the results from CEEMDAN and EEMD models. One distinguishing feature of CEEMD compared with EEMD and CEEMDAN was its superior accuracy in predicting ET_0 for the 15-day forecasting period, where CEEMD-CNN exhibited higher predictive precision. Regarding the comprehensive CA metric, CEEMD-CNN's performance surpassed that of CEEMDAN-CNN and EEMD-CNN by approximately 16.09 and 17.71%, respectively. Nonetheless, the overall predictive performance of CEEMD remained comparable to that of CEEMDAN and EEMD. Compared with the baseline model, CEEMD still demonstrated a noticeable gap in predictive performance for multi-period ET_0 prediction in humid regions.

This study conducted a detailed boxplot visualization analysis (Figure 11) of the prediction error evaluation metric PAE for ET_0 forecasts at Nanchang Station over 1- and 15-day forecasting periods. The analysis compared the baseline model (single

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Table 10 | Statistical values of the CEEMD model for forecasting ET_0 over different periods at the Nanchang station

Model	Statistical Indicators					Statistical Indicators				
ET_0 Period	R^2	MAE $mm\ d^-$	$RMSE$ $mm\ d^-$	NSE	CA	R^2	MAE $mm\ 3d$	$RMSE$ $mm\ 3d$	NSE	CA
	1-day					3-day				
CEEMD-XGB	**0.8245**	**0.5287**	**0.7239**	**0.8245**	**0.4713**	0.7429	0.5855	0.8181	0.7429	0.5480
CEEMD-SVM	0.8069	0.5893	0.7594	0.8069	0.5088	**0.7436**	**0.6300**	**0.8169**	**0.7436**	**0.5621**
CEEMD-RF	0.8079	0.5578	0.7575	0.8079	0.4974	0.7383	0.6079	0.8252	0.7383	0.5593
CEEMD-CNN	0.8272	0.5323	0.7183	0.8272	0.4697	0.7893	0.5484	0.7406	0.7893	0.4949
CEEMD-LSTM	0.8200	0.5427	0.7333	0.8200	0.4805	0.7777	0.5535	0.7606	0.7777	0.5070
CEEMD-Bi-LSTM	0.8328	0.5261	0.7066	0.8328	0.4620	0.7760	0.5601	0.7636	0.7760	0.5108
CEEMD-CNN-LSTM	0.8358	0.5154	0.7004	0.8358	0.4554	0.7777	0.5618	0.7607	0.7777	0.5098
CEEMD-CNN-Bi-LSTM	0.8142	0.5444	0.7450	0.8142	0.4868	0.7713	0.5698	0.7715	0.7713	0.5181
	5-day unit $mm\ 5d^{-1}$					7-day unit $mm\ 7d^{-1}$				
CEEMD-XGB	**0.7580**	**0.5712**	**0.7594**	**0.7580**	**0.5190**	0.6944	0.6291	0.8385	0.6944	0.5852
CEEMD -SVM	0.7591	0.5894	0.7576	0.7591	0.5240	**0.7220**	**0.6345**	**0.7998**	**0.7220**	**0.5651**
CEEMD -RF	0.7592	0.5770	0.7575	0.7592	0.5199	0.7023	0.6329	0.8277	0.7023	0.5803
CEEMD -CNN	0.7875	0.5402	0.7116	0.7875	0.4832	0.7633	0.5725	0.7379	0.7633	0.5105
CEEMD -LSTM	0.7877	0.5346	0.7113	0.7877	0.4812	0.7674	0.5625	0.7315	0.7674	0.5038
CEEMD-Bi-LSTM	0.7857	0.5373	0.7146	0.7857	0.4839	0.7544	0.5778	0.7518	0.7544	0.5198
CEEMD-CNN-LSTM	0.7502	0.5836	0.7716	0.7502	0.5297	0.7109	0.6202	0.8156	0.7109	0.5692
CEEMD -CNN-Bi-LSTM	0.7912	0.5373	0.7054	0.7912	0.4790	0.7244	0.6095	0.7964	0.7244	0.5549
	10-day unit $mm\ 10d^{-1}$					15-day unit $mm\ 15d^{-1}$				
CEEMD-XGB	0.7116	0.5821	0.7967	0.7116	0.5502	0.7116	0.5821	0.7967	0.7116	0.5502
CEEMD-SVM	**0.7394**	**0.5963**	**0.7573**	**0.7394**	**0.5327**	**0.7647**	**0.5257**	**0.6977**	**0.7647**	**0.4814**
CEEMD-RF	0.7232	0.5844	0.7805	0.7232	0.5417	0.7467	0.5200	0.7238	0.7467	0.4941
CEEMD-CNN	0.8290	0.4967	0.6134	0.8290	0.4228	0.9038	0.3477	0.4461	0.9038	0.2937
CEEMD-LSTM	0.8485	0.4424	0.5774	0.8485	0.3865	0.8753	0.4049	0.5079	0.8753	0.3424
CEEMD-Bi-LSTM	0.7854	0.5278	0.6872	0.7854	0.4718	0.8797	0.3787	0.4989	0.8797	0.3293
CEEMD-CNN-LSTM	0.7468	0.5661	0.7465	0.7468	0.5167	0.7700	0.5160	0.6898	0.7700	0.4738
CEEMD-CNN-Bi-LSTM	0.7555	0.5660	0.7335	0.7555	0.5095	0.7779	0.4996	0.6779	0.7779	0.4619

Statistical values of the CEEMDAN model for forecasting ET_0 over different periods at the Xinjian station In the table, the cells with a yellow background represent the best forecasting results, while the text in bold indicates the best forecasting results among the three conventional ML models.

model), the CEEMDAN-based hybrid model, the EEMD-based hybrid model, and the CEEMD-based hybrid model. From Figure 11(a), it is evident that the prediction errors of the CEEMDAN, EEMD, and CEEMD methods for the 1-day ET_0 forecast exceed those of the baseline model. Additionally, the three decomposition methods demonstrate similar predictive performance without significant differences. Figure 11(a) also highlights the superior performance of DL methods in multi-period ET_0 forecasting in humid regions. Similarly, for the 15-day ET_0 forecast, as shown in Figure 11(b), the predictive performance of the CEEMDAN, EEMD, and CEEMD-based hybrid models is lower than that of the baseline model. Compared with the 1-day ET_0 forecast, the combination of decomposition methods with DL models (LSTM and Bi-LSTM) shows a more pronounced advantage over traditional ML models (XGB, SVM, and RF) in terms of prediction error. This underscores the superior predictive performance of DL methods.

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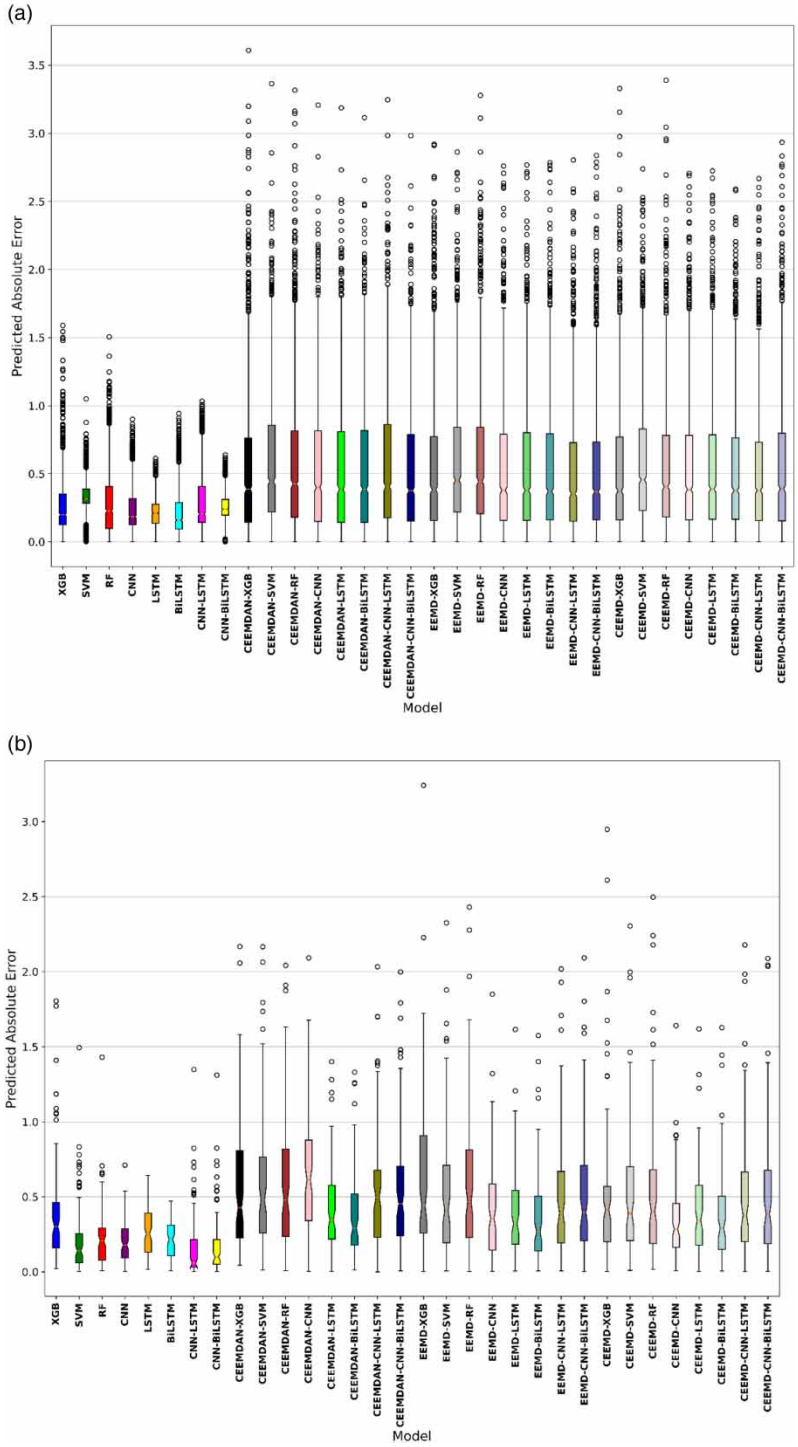

Figure 11 | Box plots of absolute forecast errors between the base model and the CEEMDAN, EEMD, and CEEMD models at Nanchang station for periods of 1 day (a) and 15 days (b).

The error analysis presented in Figure 11 clearly illustrates that the CEEMDAN, EEMD, and CEEMD methods exhibit significant performance gaps compared with the baseline model in multi-period ET_0 forecasting in humid regions. This finding further highlights the influence and limitations of decomposition methods in predicting ET_0 over multiple periods in humid environments.

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3.4. Comparison of the computational costs of EEMD, CEEMD, CEEMDAN, and baseline models

Figure 12 illustrates the computational time cost for CEEMDAN models compared with the base models across all meteorological stations. The specific computational cost unit is based on the sum of the time spent on forecasting for all periods at each station. It is evident from Figure 12 that the base CEEMDAN models incur significantly higher computational costs compared with the base models. The figure provides a detailed analysis of the time cost multiples for various models and the runtime of the superior models. The traditional ML models, XGB, SVM, and RF, exhibit extremely fast runtime at each station. This is attributed to the inherent speed of traditional ML, which lacks the complex neural network structures found in DL models. Taking station 58606 (Nanchang station) as a reference, the computational cost of CEEMDAN-based models, LSTM and Bi-LSTM, is approximately 8.9 and 8.5 times higher than that of the base LSTM and Bi-LSTM models, respectively. Similar results are observed at the other two stations. LSTM and Bi-LSTM, compared with CEEMDAN-LSTM and CEEMDAN-Bi-LSTM algorithms, have more superficial structures and lower computational costs while achieving higher forecasting accuracy. Additionally, in terms of the runtime of the EEMD model, the EEMD method required more computation time compared with CEEMDAN. Specifically, during the operation of the EEMD-Bi-LSTM model, the runtime cost was approximately 2.63 times higher than that of CEEMDAN-Bi-LSTM, with a runtime of 11,288 s. Due to the high computational cost, and to minimize the impact on forecast accuracy, this study selected the same model parameters for EEMD-CNN-LSTM and EEMD-CNN-Bi-LSTM as those used in the CEEMDAN-CNN-LSTM and CEEMDAN-CNN-Bi-LSTM models, which reduced the runtime by approximately five times. As shown in Figure 12, CEEMD still incurs significantly higher runtime costs compared with CEEMDAN. Furthermore, the EEMD and CEEMD methods with traditional ML models incur lower computational costs than CEEMDAN. In comparison with the EEMD, CEEMD, and CEEMDAN-LSTM and EEMD, CEEMD, and CEEMDAN-Bi-LSTM algorithms, both LSTM and Bi-LSTM models exhibit a shallower structure and lower computational cost while achieving higher prediction accuracy. As illustrated in Figure 12, the CNN-Bi-LSTM model based on CEEMDAN demonstrates significantly higher

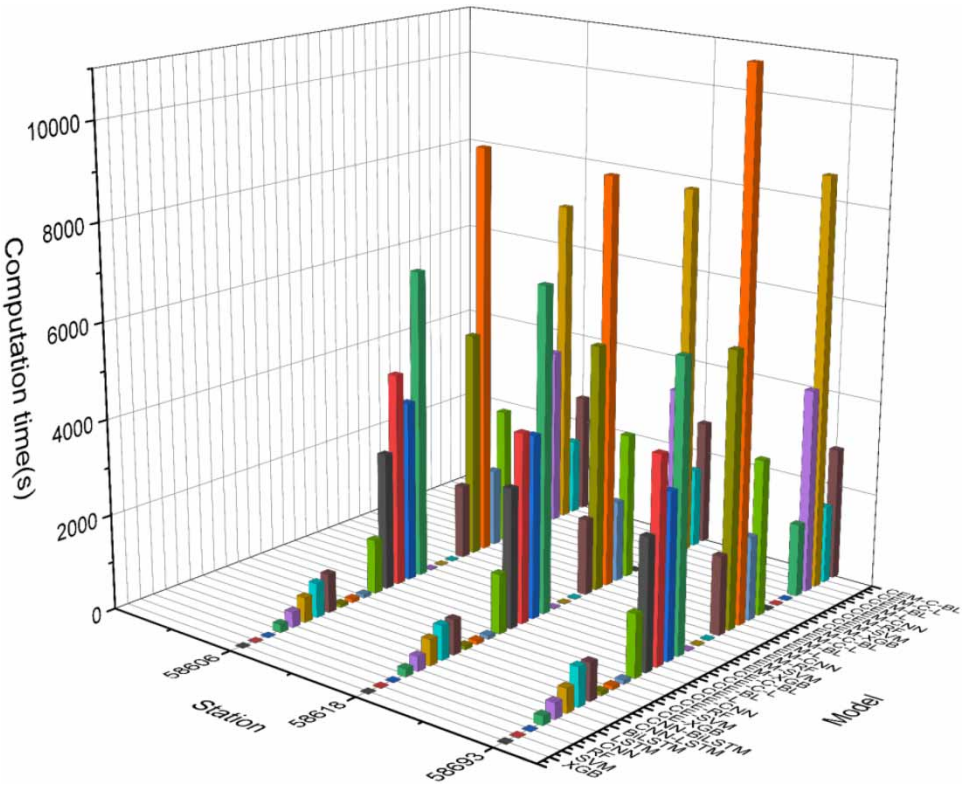

Figure 12 | Three-dimensional histograms of the forecasting time cost for different ET_0 periods by the base models and EEMD, CEEMD, and CEEMDAN models at Nanchang, Dongxiang, and Xinjian stations. Note that in the figure, CE represents CEEMDA, EM represents EEMD, CEM represents CEEMD, C represents CNN, L represents LSTM, and BL represents Bi-LSTM.

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runtime costs, highlighting the superior efficiency of LSTM and Bi-LSTM algorithms and the disadvantages of using the EEMD, CEEMD, and CEEMDAN methods.

4. DISCUSSION

4.1. Strategies for forecasting ET_0 in different periods

Accurate forecasting of ET_0 at different periods is crucial for addressing practical issues, and the model's accuracy varies significantly for different ET_0 periods. The results of this study indicate that both LSTM and Bi-LSTM models achieve satisfactory results in forecasting ET_0 at different periods. However, there are some inconsistencies, and as the forecasting period lengthens, the forecasting accuracy tends to decrease. Lu *et al.* (2023) used VMD-BPNN to forecast ET_0 at different periods and found that the forecasting accuracy decreases as the forecasting period lengthens. This result is similar to the patterns observed in this study's forecasting of CNN and XGB models. However, in contrast, the forecasting accuracy of LSTM and Bi-LSTM in this study is less affected by the length of the forecasting period and consistently achieves good results across all periods. Multi-step forward forecasting and nonperiodic forecasting of ET_0 have different meanings. Multi-step forward forecasting allows for advanced knowledge and understanding of the specific data of ET_0. Malik *et al.* (2022) used the MVMD-RR technique to optimize the KELM model for accurate forecasting of ET_0 for the next 3 and 7 days. However, there is still a phenomenon of decreasing forecasting accuracy as the forecasting length increases. Compared with this experiment, forecasting ET_0 for different periods allows for a more precise understanding of the specific trends in each period. The results indicate that LSTM and Bi-LSTM do not exhibit a linear decline in forecasting accuracy with changes in the forecasting period length.

4.2. Comparison of forecasting accuracy between EEMD, CEEMD, CEEMDAN, and the baseline model CEEMDAN

The basic DL models, LSTM and Bi-LSTM, exhibited similar performance in forecasting ET_0 for different periods across all stations, outperforming other base models and CEEMDAN models. Mandal & Chanda (2023) found that LSTM had higher forecasting accuracy than SVR, RF, MLP, and MARS in real-time and 28-day ahead ET_0 forecasting using DL. Ayaz *et al.* (2021) estimated ET_0 in two different climatic regions using ML. The evaluation of results indicated that LSTM models outperformed GBR, SVR, and RF models. Long *et al.* (2022) decomposed the original data through a DWT decomposition model and used LSTM, ANN, and ELM models as base models to estimate ET_0 in Beijing and Baoding in the northern part of the North China Plain. The results indicated that single ML models (LSTM, ANN, and ELM) were more stable than models based on DWT (wavelet-coupled) decomposition, and they generally exhibited better overall performance. It is worth noting that in a considerable amount of previous research on estimating and forecasting ET_0, good results have been obtained using non-CEEMDAN decomposition models. For example, Kang *et al.* (2022) achieved satisfactory results in forecasting ET_0 at 10 stations in the Weihe River Basin in China by combining VMD and Box–Cox transformation (BC) with SVM. Jayasinghe *et al.* (2021) demonstrated the utility of combining MEMD and Boruta with LSTM in forecasting daily ET_0 datasets, showing its usefulness compared with independent LSTM models. Yan *et al.* (2023) achieved excellent water-level forecasting results by combining the CEEMDAN decomposition model with LSTM and Bi-LSTM models. The specific analysis of the experimental results above is shown in Table 11. From Table 11, in previous studies, the combination of decomposition algorithms and ML models has yielded satisfactory results for various forecasting tasks. However, in this experiment, the opposite forecasting effect was obtained. In this study, the addition of EEMD, CEEMD, and CEEMDAN led to a decrease in prediction accuracy compared with the base model. The reason for this phenomenon is that the incorporation of EEMD, CEEMD, and CEEMDAN generates components for each feature in the original data. LSTM accepts different component features within a fixed parameter range, greatly increasing the difficulty of generalizing the LSTM model. This significantly increases the dependence on parameters of LSTM and other models, resulting in a significant decrease in forecasting accuracy with parameters that are not suitable for component features, ultimately leading to lower forecasting results than the base model. Additionally, due to the high quality of the data in this experiment, even without using a decomposition algorithm to denoise the raw data, decompose the original signal with higher nonlinearity, and then sequentially forecast each IMF mode, the base model still maintains a high level of forecasting accuracy. The basic LSTM and Bi-LSTM models achieved the best results in forecasting ET_0 for different periods in this experiment. However, models based on EEMD, CEEMD, and CEEMDAN have lower prediction accuracy than the base models. This is closely related to the data decomposition by EEMD, CEEMD, and CEEMDAN, which introduce changes in input features, requiring the derivation of optimal model parameters corresponding to the altered data. This complexity

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Table 11 | Literature on the combination of decomposition algorithms and ML

Author/ References	Research objective	Best model used	Performance of models
Long *et al.* (2022)	Reference evapotranspiration estimation using long short-term memory network and wavelet-coupled long short-term memory network	Wavelet-coupled-LSTM	$R^2 = 0.998$, RMSE = 0.08 mm day^{-1}, MAE = 0.059 mm day^{-1}, NSE = 0.997, when Rn, T, RH, U is input.
Kang *et al.* (2022)	Machine learning framework with decomposition–transformation and identification of key modes for estimating reference evapotranspiration	VMD-Box–Cox (BC)-SVM	The correlation coefficient (R) is consistently greater than 0.96; both MAPE (mean absolute percentage error) and RMSE are less than 8.41% and 0.38 mm day^{-1}, respectively.
Jayasinghe *et al.* (2021)	Deep multi-stage reference evapotranspiration forecasting model	MEMD-Boruta-LSTM	$R = 0.9668$, RMSE = 0.5307 mm day^{-1}, MAE = 0.4204 mm day^{-1} s, NSE = 0.8960.
Yan *et al.* (2023)	Forecasting water levels using CEEMDAN decomposition algorithm with machine learning and deep learning ensemble	CEEMDAN-Bi-LSTM	$R^2 = 0.9846$, MAE = 0.501 m h^{-1}, RMSE = 0.0641 m h^{-1}, nRMSE = 0.0235.

suggests that EEMD, CEEMD, and CEEMDAN decomposition methods may not be suitable for predicting multi-period ET_0 forecasts in humid regions ET_0.

4.3. Stability of the EEMD, CEEMD, CEEMDAN, and base models

The stability of the model is an important factor to consider when forecasting reliable ET_0. In this study, the three conventional ML models, XGB, SVM, and RF, exhibit the highest percentage variations in R^2, RMSE, MAE, NSE, and CA in forecasting ET_0 for different periods across all stations. Among them, XGB shows the most noticeable variation in RMSE in forecasting ET_0 for different periods. This phenomenon also reveals that ML-based XGB, SVM, and RF models, when forecasting different periods of ET_0, exhibit a decrease in forecasting performance as the period changes, leading to a reduction in stability. However, for models based on CEEMDAN, judging stability based on the increase in RMSE as a criterion for stability in forecasting different periods of ET_0 is more stable than the base model. However, for models based on EEMD, CEEMD, and CEEMDAN, the increase in RMSE and CA metrics serves as a criterion for evaluating the stability of ET_0 predictions across different time periods, revealing that their stability is lower than that of the baseline model. Nevertheless, models based on EEMD, CEEMD, and CEEMDAN should be considered as a cohesive system rather than being evaluated against the stability criteria of the baseline model. These models fail to achieve optimal solutions for different stations and time periods. Figure 10 clearly demonstrates that models based on EEMD, CEEMD, and CEEMDAN demonstrate weaker generalization capabilities, making them incapable of adapting to changes in input features. Consequently, these models produce inconsistent predictions and fail to reliably forecast ET_0 variations. In contrast, the base models, LSTM and Bi-LSTM, obtained similar forecasting results for ET_0 at different stations and periods. The percentage increase in RMSE for these models is smaller than for all models in this experiment. Moreover, Figure 12 also shows that the baseline model has the best overall performance compared with the EEMD, CEEMD, and CEEMDAN decomposition models. It also responds more steadily to changes in features, making ET_0 predictions that are consistent and accurate. Additionally, LSTM and Bi-LSTM based on DL have a large number of parameters that need to be adjusted in real-time based on changes in input features. Therefore, further parameter optimization using GS methods is needed to select the best model parameters to ensure the excellent stability of the models.

4.4. Computational costs of the EEMD, CEEMD, CEEMDAN, and base models

The results indicate that the runtime of models based on CEEMDAN is significantly higher than the corresponding base models. Specifically, at the Nanchang station, the computational cost of the CEEMDAN-LSTM model is approximately 8.9 times that of the base LSTM model, and the computational cost of the CEEMDAN-Bi-LSTM model is about 8.5 times that of the base Bi-LSTM model. However, the high multiple of runtime costs comes at the expense of an inversely proportional impact on the final forecasting accuracy. Models based on CEEMDAN exhibit very low operational efficiency. Moreover, the runtime costs for models based on CEEMDAN and CNN are even higher for LSTM and Bi-LSTM. The

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computational costs of the three conventional ML models, XGB, RF, and SVM, are significantly lower than those of various DL models. However, their forecasting accuracy and stability are lower than those of LSTM and Bi-LSTM. The computational costs of ML increase with the values and quantity of the model parameters used (Fan *et al.* 2018). DL shares similar principles with ML. LSTM and Bi-LSTM, based on DL, have many parameters, leading to higher computational costs than ML. However, obtaining better forecasting results through higher computational time costs is acceptable. In this experiment, the base models were more efficient in computation time than the CEEMDAN algorithm. Combining high-precision forecasting by LSTM and Bi-LSTM with lower computational time costs than CEEMDAN makes them better models for forecasting ET_0. CEEMDAN, which is effective for complex nonlinear problems, decomposes each variable into complex-to-simple signals through multiple iterations and aggregates them to produce a forecast. However, in this study, using CEEMDAN did not improve ET_0 forecasting performance. This is likely due to the strong correlation within the ET_0 data itself. For instance, the regional limitations of the forecast area cause significant periodic variations in ET_0. This might lead to substantial learning errors when CEEMDAN decomposes signals into simpler components, resulting in inaccurate forecasts. Additionally, decomposing multiple variables with CEEMDAN and reusing them as model inputs results in forecasts equivalent to those based on different datasets, causing a loss of accuracy due to the inconsistency of inputs.

Additionally, we conducted a detailed analysis of the computational time costs for the EEMD and CEEMD models. The results indicate that the EEMD model, when combined with traditional ML methods, demonstrates lower computational time costs compared with the CEEMDAN model. However, this difference is negligible when comparing the computational costs of EEMD and CEEMD combined with DL models. For instance, in the case of EEMD combined with the Bi-LSTM model, the computational cost exceeds 10,000 s, which is 2.6 times higher than that of CEEMDAN-Bi-LSTM. Furthermore, the computational cost of the decomposition methods shows a substantial increase compared with the baseline model. To address this, in this study, EEMD and CEEMD combined with CNN-LSTM and CNN-Bi-LSTM underwent model parameter adjustments. To minimize the impact on forecast accuracy, the model parameters were set to be identical to those of CEEMDAN-CNN-LSTM and CEEMDAN-CNN-Bi-LSTM. Experimental results show that under these parameter settings, computational time was reduced by a factor of five.

However, even without the need for hyperparameter GS algorithms, the computational costs of EEMD and CEEMD models combined with CNN-LSTM and CNN-Bi-LSTM under a single parameter setting still approached half an hour. When using the same GS algorithms as the CEEMDAN and baseline models, the computational time costs for EEMD and CEEMD combined with CNN-LSTM and CNN-Bi-LSTM models remained significantly higher than those of the baseline and CEEMDAN models. This study recommends using LSTM and Bi-LSTM as forecasting models for ET_0 in future studies of humid regions. Additionally, it provides valuable insights into the use of decomposition models combined with ML for ET_0 forecasting.

In this study, we compared the performance of LSTM and Bi-LSTM models with other models in forecasting ET_0 across different periods. The results indicate that the LSTM and Bi-LSTM models demonstrate high accuracy and stability in predicting ET_0, consistent with the findings of Mandal & Chanda (2023), who observed that LSTM showed greater forecasting accuracy than SVR, RF, MLP, and MARS in ET_0 prediction. Additionally, the research by Ayaz *et al.* (2021) also indicated that the LSTM model outperformed the GBR, SVR, and RF models in estimating ET_0 in two distinct climate regions. Moreover, a key finding of this study is that EEMD, CEEMD, and CEEMDAN did not outperform the basic LSTM and Bi-LSTM models in multi-period ET_0 prediction for humid regions of China. This could be due to changes in the correlation between ET_0-decomposed IMFs and corresponding IMFs of other meteorological variables after decomposition, potentially weakening these correlations. Furthermore, the temporal dependencies in time-series data may be affected, limiting the model's ability to capture the time-dependent nature of the data accurately, thus leading to a decline in ET_0 prediction accuracy. It was also observed that these decomposition methods generate multiple IMFs, requiring repetitive predictions for each IMF followed by an aggregation process, which significantly increases the computational cost of the EEMD, CEEMD, and CEEMDAN models. In conclusion, this study highlights the potential of EEMD, CEEMD, and CEEMDAN decomposition methods for multi-period ET_0 prediction in humid regions of China while emphasizing that ET_0 prediction should consider not only model accuracy but also computational cost and stability. Due to their superior performance in these aspects, the LSTM and Bi-LSTM models are recommended for multi-period ET_0 forecasting research in humid regions of China. Our research also provides valuable insights and recommendations for the application of ML combined with decomposition models in ET_0 forecasting.

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5. CONCLUSIONS

In this study, three conventional ML models, XGB, RF, SVM, and DL models, including CNN, LSTM, Bi-LSTM, CNN-LSTM, and CNN-Bi-LSTM, were utilized. In addition, a hybrid model that combines EEMD, CEEMD, and CEEMDAN with the aforementioned models, respectively, was introduced. The objective was to evaluate the accuracy and stability of forecasting different periods of ET_0. In the forecasting of ET_0 for the Nanchang station, the base models, LSTM ($R^2 = 0.968$, MAE = 0.2245 mm 5day^{-1}, RMSE = 0.2686mm 5day^{-1}, NSE = 0.968, CA = 0.1727) and Bi-LSTM ($R^2 = 0.9768$, MAE = 0.2016mm day^{-1}, RMSE = 0.2630mm day^{-1}, NSE = 0.9768, CA = 0.1610), achieved satisfactory results. For this station, the best-performing model within the CEEMDAN-based hybrid framework is CEEMDAN-Bi-LSTM ($R^2 = 0.8872$, MAE = 0.4853mm 15day^{-1}, RMSE = 0.3863 mm 15day^{-1}, NSE = 0.8872, CA = 0.3249). Within the EEMD-based hybrid framework, the best-performing model is CEEMD-Bi-LSTM ($R^2 = 0.8905$, MAE = 0.3606mm 15day^{-1}, RMSE = 0.4759 mm 15day^{-1}, NSE = 0.8905, CA = 0.3122). Meanwhile, the best-performing model within the CEEMD-based hybrid framework is CEEMD-CNN ($R^2 = 0.9038$, MAE = 0.3477 mm 15day^{-1}, RMSE = 0.4461 mm 15day^{-1}, NSE = 0.9038, CA = 0.2937). However, compared with the best-performing model (Bi-LSTM) for the 1-day ET_0 forecast at the Nanchang station, the EEMD-, CEEMD-, and CEEMDAN-based hybrid Bi-LSTM models exhibited significantly lower forecasting performance in terms of the comprehensive metric CA for the 1-day ET_0 forecast. Specifically, the prediction performances of the three decomposition models were approximately 193.30, 187.02, and 191.91% lower than that of the baseline Bi-LSTM model, respectively. Furthermore, a statistical analysis of the best models for different stations reveals that the base models consistently obtained a more unified best-performing model for ET_0 forecasting across various stations and periods. However, the CEEMDAN-based hybrid models exhibited inconsistent best models for forecasting ET_0 across different stations and periods. This indicates that introducing the CEEMDAN model resulted in poorer forecasting stability. Surprisingly, the analysis of model computational costs revealed an unacceptable increase in the running costs for each CEEMDAN-based hybrid model. The base model Bi-LSTM achieved satisfactory results in forecasting different stations, with an average runtime cost of around 8 min across the three stations. However, after incorporating CEEMDAN, the runtime cost for CEEMDAN-Bi-LSTM increased to approximately 1 h and 10 min. This represents an increase of about 8–9 times in runtime costs after adding CEEMDAN, which is unacceptable. Furthermore, both EEMD and CEEMD demonstrated even higher computational costs compared with CEEMDAN while still showing lower stability than the baseline model, with the computational cost of EEMD-Bi-LSTM being 2.6 times that of CEEMDAN-Bi-LSTM. In conclusion, based on the experimental results, the EEMD, CEEMD, and CEEMDAN models are not suitable for forecasting ET_0 over different time periods. It is worth noting that incorporating LSTM and Bi-LSTM with CNN is also a decision that requires attention. In this experiment, adding CNN to LSTM and Bi-LSTM did not yield good forecasting results and increased runtime costs. The probable reason for this phenomenon is that the research data in this experiment has high quality, and excessive preprocessing, further feature extraction, and noise reduction methods may have had a counterproductive effect. Similar to enhancing positive relationships but diminishing negative relationships, this could have led to poorer forecasting outcomes. This study investigated the use of EEMD, CEEMD, and CEEMDAN decomposition methods combined with ML to predict ET_0 in humid regions. The results indicated that the EEMD, CEEMD, and CEEMDAN hybrid models did not outperform the baseline model. It is recommended to use LSTM and Bi-LSTM models for forecasting ET_0 at different periods in humid regions. Based on the findings, caution should be exercised when using EEMD, CEEMD, or CEEMDAN combined with ML for ET_0 prediction. Given their accuracy and computational costs, EEMD, CEEMD, or CEEMDAN combined with ML are not optimal choices for ET_0 forecasting in humid regions and are therefore not recommended for this purpose. Conversely, single models such as XGB, RF, and SVM can be considered for ET_0 forecasting in humid regions due to their accuracy and time efficiency. It is important to note that this study has some limitations. The integration of EEMD, CEEMD, or CEEMDAN with DL did not show significant improvement in ET_0 prediction and demonstrated unstable performance across multiple stations and time periods. Additionally, the introduction of EEMD, CEEMD, and CEEMDAN significantly increased the computational cost of the models. In this research, LSTM and Bi-LSTM achieved precise multi-period ET_0 forecasts for humid regions in China, but they were not validated in other regions, so it cannot be ensured that similar accuracy would be achieved in those areas. In future studies, ET_0 forecasting for more regions will be conducted to verify the accuracy and stability of LSTM and Bi-LSTM models for multi-period ET_0 forecasting across different regions.

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ACKNOWLEDGEMENTS

We would like to express our special thanks to the National Meteorological Information Center of China Meteorological Administration for providing the foundational data for this research.

AUTHOR CONTRIBUTIONS

Conceptualization, Z.Y. and L.W.; methodology, Z.Y. and L.W.; software, Z.Y.; validation, Z.Y., L.W., and X.L.; formal analysis, Z.Y. and L.W.; investigation, Z.Y., L.W., and X.L.; resources, Z.Y. and L.W.; data curation, Z.Y.; writing – original draft preparation, Z.Y.; writing – review and editing, Z.Y. and L.W.; visualization, Z.Y. and L.W.; supervision, Z.Y., L.W., and X.L.; project administration, X.L.; funding acquisition, X.L. All authors have read and agreed to the published version of the manuscript.

FUNDING

This research was funded by the General Project of Natural Science Foundation of Jiangxi Province (Grant No. 20232BAB205031), the National Natural Science Foundation of China (Grant No. 52269013), and the Key Project of Natural Science Foundation of Jiangxi Province (Grant No. 20242BAB26081).

DATA AVAILABILITY STATEMENT

Data cannot be made publicly available; readers should contact the corresponding author for details.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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First received 27 May 2024; accepted in revised form 4 February 2025. Available online 17 February 2025

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